

Lesson 2 – Engineering Drawings and Tolerancing**Engineering Drawing:**

An engineering drawing documents an architecture or an engineering design.

A blueprint is a paper reproduction of that drawing.

In the machining industry it details the design of machined or manufactured parts.

Generally, "blueprint" is the term used that refers to any detailed plan.

Tolerance:

The amount of variation permitted on the surfaces of manufactured parts

It is impossible to manufacture to exact sizes, so tolerances are specified in the title block. These are the most important drawing aspects.

STANDARD TOLERANCES	
UNLESS OTHERWISE SPECIFIED	
INCH UNITS	
FRACTIONAL	+/- 1/64
2 PLC. DECIMAL	+/- 0.01
3 PLC. DECIMAL	+/- 0.005
4 PLC. DECIMAL	+/- 0.0005
ANGLES	+/- 1/2°
DEBURR ALL SHARP EDGES	

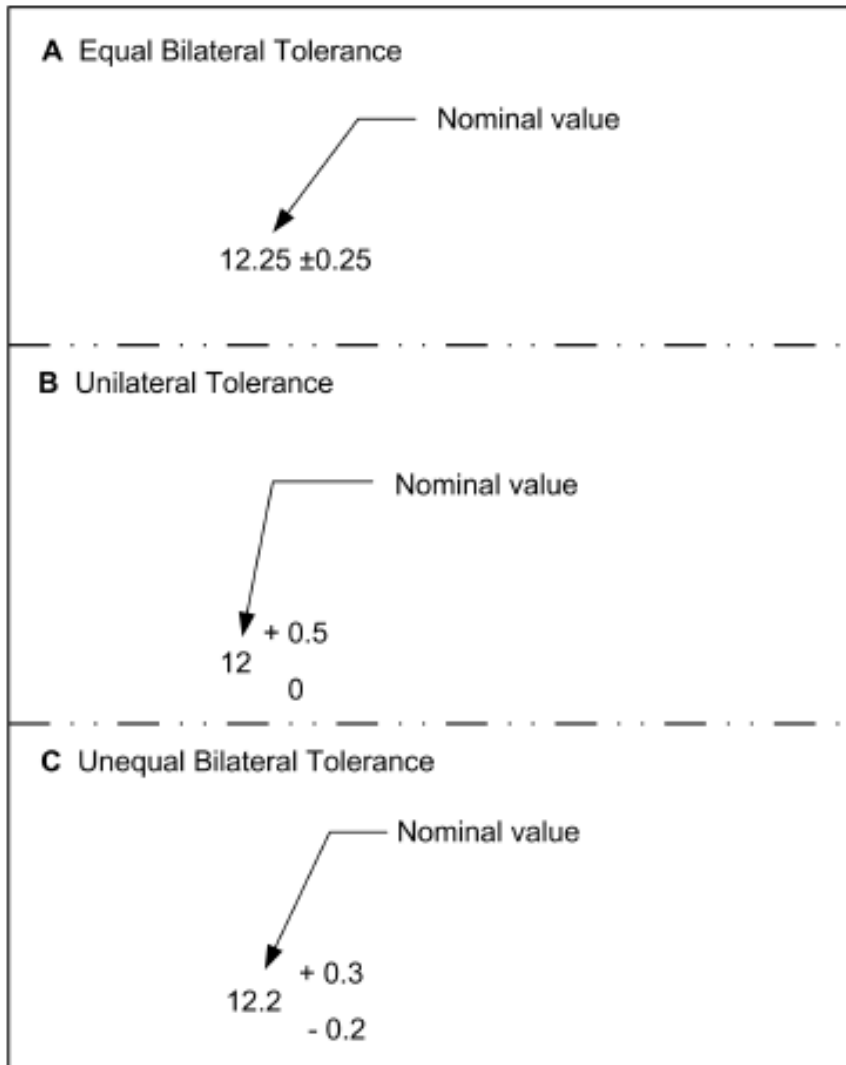
Types of Tolerances:

Two common types, *Limit Tolerances* and *Plus-Minus Tolerances*

LIMITS: the extreme permissible dimensions of part sizes or measurements

LIMIT TOLERANCE: when a dimension has high and low limits stated

PLUS-MINUS TOLERANCE: dimension is given as a nominal, target value, followed by a $\pm$ expression



Fits of Mating Parts

Have wide application in the manufacturing industry, for example shafts and threads.
Tolerance applied determines the looseness or tightness of fit between parts.

Clearance Fit:

When parts are designed to move with one another, there is space (or, clearance) between the moving parts

-i.e. a shaft turning in a bushing. The shaft diameter must be smaller than the bushing diameter

Interference Fit:

When parts are designed to force fit within another, interference exists

-i.e. a pin pressed into a hole. The pin diameter is greater than the hole diameter

Maximum Clearance equals: -max hole diameter, min shaft diameter

Minimum Clearance equals: -min hole diameter, max shaft diameter

Maximum Interference equals: -max pin diameter, min hole diameter

Minimum Interference equals: -min pin diameter, max hole diameter

Example of a Clearance Fit:

-eg. a shaft and hole with clearance fit and bilateral tolerances

$$\text{Max Shaft Diameter} = 2.7502'' + 0.0008'' = 2.751''$$

$$\text{Min Shaft Diameter} = 2.7502'' - 0.0008'' = 2.7494''$$

$$\text{Max Hole Diameter} = 2.7536'' + 0.0008'' = 2.7544''$$

$$\text{Min Hole Diameter} = 2.7536'' - 0.0008'' = 2.7528''$$

Max Clearance

$$= \text{Max Hole Diameter} - \text{Min Shaft Diameter}$$

$$= 2.7544'' - 2.7494''$$

$$= 0.0050''$$

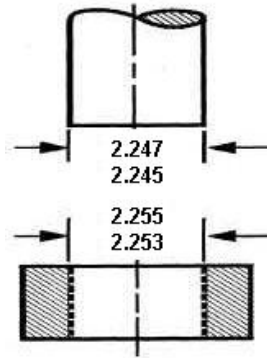
Min Clearance

$$= \text{Min Hole Diameter} - \text{Max Shaft Diameter}$$

$$= 2.7528'' - 2.751''$$

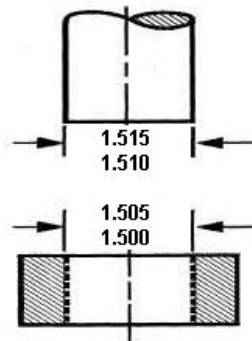
$$= -0.0018''$$

Calculate the maximum and minimum clearances between the following two mating parts.



Maximum Clearance: _____ Minimum Clearance: _____

Calculate the maximum and minimum interferences between the following two mating parts.



Maximum Interference: _____ Minimum Interference: _____

ANSI Standard Fits and Tolerances:

ANSI Standard Limits and Fits (ANSI B4.1-1967 (R1994)). This American National Standard for Preferred Limits and Fits presents definitions of terms applying to fits between cylindrical parts and makes recommendations on preferred sizes, allowances, tolerances, and fits for use wherever they are applicable.

- This standard is up to a diameter of 20 inches.
- As revised in 1967, and reaffirmed in 1979, the definitions in ANSI B4.1 have been expanded and some of the limits in certain classes have been changed.

Definitions.: The following terms are defined in this standard:

Nominal Size: The nominal size is the designation used for the purpose of general identification.

Allowance: An allowance is a prescribed difference between the maximum material limits of mating parts. It is a minimum clearance (positive allowance), or maximum interference (negative allowance) between such parts.

Tolerance: A tolerance is the total permissible variation of a size.

Basic Size: The basic size is that size from which the limits of size are derived by the application of allowances and tolerances.

Fit: Fit is the general term used to signify the range of tightness that may result from the application of a specific combination of allowances and tolerances in the design of mating parts.

Actual Fit: The actual fit between two mating parts is the relation existing between them with respect to the amount of clearance or interference that is present when they are assembled.

Clearance Fit: A clearance fit is one having limits of size so specified that a clearance always results when mating parts are assembled.

Interference Fit: An interference fit is one having limits of size so specified that an interference always results when mating parts are assembled.

Transition Fit: A transition fit is one having limits of size so specified that either a clearance or an interference may result when mating parts are assembled.

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Basic Hole System: A basic hole system is a system of fits in which the design size of the hole is the basic size and the allowance, if any, is applied to the shaft.

Basic Shaft System: A basic shaft system is a system of fits in which the design size of the shaft is the basic size and the allowance, if any, is applied to the hole.

Standard Tolerances:

- The term “*tolerance grade*” refers to a set of tolerances that can be produced with an approximately equal production capability.
- The actual total tolerance allowed within each grade depends on the nominal size of the dimension.
- Smaller tolerances can be achieved in smaller dimensions, and vice versa.

Table 1. ANSI Standard Tolerances ANSI B4.1-1967 (R1987)

Nominal Size, Inches		Grade									
		4	5	6	7	8	9	10	11	12	13
Over	To	Tolerances in thousandths of an inch ^a									
0	0.12	0.12	0.15	0.25	0.4	0.6	1.0	1.6	2.5	4	6
0.12	0.24	0.15	0.20	0.3	0.5	0.7	1.2	1.8	3.0	5	7
0.24	0.40	0.15	0.25	0.4	0.6	0.9	1.4	2.2	3.5	6	9
0.40	0.71	0.2	0.3	0.4	0.7	1.0	1.6	2.8	4.0	7	10
0.71	1.19	0.25	0.4	0.5	0.8	1.2	2.0	3.5	5.0	8	12
1.19	1.97	0.3	0.4	0.6	1.0	1.6	2.5	4.0	6	10	16
1.97	3.15	0.3	0.5	0.7	1.2	1.8	3.0	4.5	7	12	18
3.15	4.73	0.4	0.6	0.9	1.4	2.2	3.5	5	9	14	22
4.73	7.09	0.5	0.7	1.0	1.6	2.5	4.0	6	10	16	25
7.09	9.85	0.6	0.8	1.2	1.8	2.8	4.5	7	12	18	28
9.85	12.41	0.6	0.9	1.2	2.0	3.0	5.0	8	12	20	30
12.41	15.75	0.7	1.0	1.4	2.2	3.5	6	9	14	22	35
15.75	19.69	0.8	1.0	1.6	2.5	4	6	10	16	25	40
19.69	30.09	0.9	1.2	2.0	3	5	8	12	20	30	50
30.09	41.49	1.0	1.6	2.5	4	6	10	16	25	40	60
41.49	56.19	1.2	2.0	3	5	8	12	20	30	50	80
56.19	76.39	1.6	2.5	4	6	10	16	25	40	60	100
76.39	100.9	2.0	3	5	8	12	20	30	50	80	125
100.9	131.9	2.5	4	6	10	16	25	40	60	100	160
131.9	171.9	3	5	8	12	20	30	50	80	125	200
171.9	200	4	6	10	16	25	40	60	100	160	250

^a All tolerances above heavy line are in accordance with American-British-Canadian (ABC) agreements.

Table 2. Relation of Machining Processes to Tolerance Grades

	MACHINING OPERATION	TOLERANCE GRADES										
		4	5	6	7	8	9	10	11	12	13	
This chart may be used as a general guide to determine the machining processes that will under normal conditions, pro- duce work within the toler- ance grades indicated.	Lapping & Honing	■	■									
	Cylindrical Grinding	■	■	■	■							
	Surface Grinding	■	■	■	■	■						
	Diamond Turning	■	■	■	■	■						
	Diamond Boring	■	■	■	■	■						
	Broaching	■	■	■	■	■	■					
	Reaming			■	■	■	■	■	■			
	Turning				■	■	■	■	■	■	■	■
	Boring					■	■	■	■	■	■	■
	Milling						■	■	■	■	■	■
	Planing & Shaping							■	■	■	■	■
	Drilling								■	■	■	■

Designation of Standard Fits:

Standard fits are designated by means of the following symbols, the symbols are not intended to be shown on manufacturing drawings; instead, sizes should be specified in drawings.

The letter symbols used are as follows:

RC =Running or Sliding Clearance Fit

LC =Locational Clearance Fit

LT = Locational Transition Clearance or Interference Fit

LN =Locational Interference Fit

FN =Force or Shrink Fit

Clearance Fits:

- When there must always be a clearance between mating parts, a clearance fit is specified.
- The designation for standard clearance fits from ANSI Standard B4.1 is the running or sliding clearance fit (RC).
- Within this standard, there are 9 classes, RC1 through RC9,
 - RC1 (close sliding fit): Accurate location of parts that must assemble without perceptible play.

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- RC2 (sliding fit): Parts that will move and turn easily but are not intended to run freely. Parts may seize with small temperature changes, especially in the larger sizes
- RC3 (precision running fit): Precision parts operating at slow speeds and light loads that must run freely. Changes in temperature may cause difficulties.
- RC4 (close running fit): Accurate location with minimum play for use under moderate loads and speeds. A good choice for accurate machinery
- RC5 (medium running fit): Accurate machine parts for higher speeds and/or loads than RC4.
- RC6 (medium running fit): Similar to RC5 for applications in which larger clearance is desired.
- RC7 (free running fit): Reliable relative motion under wide temperature variations in applications where accuracy is not critical.
- RC8 (loose running fit): Permits large clearances, allowing the use of parts with commercial, “as received” tolerances
- RC9 (loose running fit): Similar to RC8, with approximately 50% larger clearances.

In summary:

- RC2 represent the precision fits (RC1, RC2, RC3); let RC5 represent the accurate, reliable running fits (RC4 to RC7); and let RC8 represent the loose fits (RC8, RC9).

The complete standard ANSI B4.1 lists the tolerances on the mating parts and the resulting limits of clearances for all 9 classes and for sizes from 0 to 200 in.

American National Standard Running and Sliding Fits ANSI B4.1-1967 (R1987)

Nominal Size Range, Inches	Class RC 1			Class RC 2			Class RC 3			Class RC 4		
	Clear- ance ^a	Standard Tolerance Limits		Clearance ^a	Standard Tolerance Limits		Clearance ^a	Standard Tolerance Limits		Clearance ^a	Standard Tolerance Limits	
		Hole H5	Shaft g4		Hole H6	Shaft g5		Hole H7	Shaft f6		Hole H8	Shaft f7
Over To	Values shown below are in thousandths of an inch											
0 – 0.12	0.1	+0.2	–0.1	0.1	+0.25	–0.1	0.3	+0.4	–0.3	0.3	+0.6	–0.3
	0.45	0	–0.25	0.55	0	–0.3	0.95	0	–0.55	1.3	0	–0.7
0.12 – 0.24	0.15	+0.2	–0.15	0.15	+0.3	–0.15	0.4	+0.5	–0.4	0.4	+0.7	–0.4
	0.5	0	–0.3	0.65	0	–0.35	1.12	0	–0.7	1.6	0	–0.9
0.24 – 0.40	0.2	+0.25	–0.2	0.2	+0.4	–0.2	0.5	+0.6	–0.5	0.5	+0.9	–0.5
	0.6	0	–0.35	0.85	0	–0.45	1.5	0	–0.9	2.0	0	–1.1
0.40 – 0.71	0.25	+0.3	–0.25	0.25	+0.4	–0.25	0.6	+0.7	–0.6	0.6	+1.0	–0.6
	0.75	0	–0.45	0.95	0	–0.55	1.7	0	–1.0	2.3	0	–1.3
0.71 – 1.19	0.3	+0.4	–0.3	0.3	+0.5	–0.3	0.8	+0.8	–0.8	0.8	+1.2	–0.8
	0.95	0	–0.55	1.2	0	–0.7	2.1	0	–1.3	2.8	0	–1.6
1.19 – 1.97	0.4	+0.4	–0.4	0.4	+0.6	–0.4	1.0	+1.0	–1.0	1.0	+1.6	–1.0
	1.1	0	–0.7	1.4	0	–0.8	2.6	0	–1.6	3.6	0	–2.0
1.97 – 3.15	0.4	+0.5	–0.4	0.4	+0.7	–0.4	1.2	+1.2	–1.2	1.2	+1.8	–1.2
	1.2	0	–0.7	1.6	0	–0.9	3.1	0	–1.9	4.2	0	–2.4
3.15 – 4.73	0.5	+0.6	–0.5	0.5	+0.9	–0.5	1.4	+1.4	–1.4	1.4	+2.2	–1.4
	1.5	0	–0.9	2.0	0	–1.1	3.7	0	–2.3	5.0	0	–2.8
4.73 – 7.09	0.6	+0.7	–0.6	0.6	+1.0	–0.6	1.6	+1.6	–1.6	1.6	+2.5	–1.6
	1.8	0	–1.1	2.3	0	–1.3	4.2	0	–2.6	5.7	0	–3.2
7.09 – 9.85	0.6	+0.8	–0.6	0.6	+1.2	–0.6	2.0	+1.8	–2.0	2.0	+2.8	–2.0
	2.0	0	–1.2	2.6	0	–1.4	5.0	0	–3.2	6.6	0	–3.8
9.85 – 12.41	0.8	+0.9	–0.8	0.8	+1.2	–0.8	2.5	+2.0	–2.5	2.5	+3.0	–2.5
	2.3	0	–1.4	2.9	0	–1.7	5.7	0	–3.7	7.5	0	–4.5
12.41 – 15.75	1.0	+1.0	–1.0	1.0	+1.4	–1.0	3.0	+2.2	–3.0	3.0	+3.5	–3.0
	2.7	0	–1.7	3.4	0	–2.0	6.6	0	–4.4	8.7	0	–5.2
15.75 – 19.69	1.2	+1.0	–1.2	1.2	+1.6	–1.2	4.0	+2.5	–4.0	4.0	+4.0	–4.0
	3.0	0	–2.0	3.8	0	–2.2	8.1	0	–5.6	10.5	0	–6.5

^aPairs of values shown represent minimum and maximum amounts of clearance resulting from application of standard tolerance limits.

Table 4. American National Standard Running and Sliding Fits ANSI B4.1-1967 (R1987)

Nominal Size Range, Inches		Class RC 5			Class RC 6			Class RC 7			Class RC 8			Class RC 9		
		Clear- ance ^a	Standard Tolerance Limits		Clear- ance ^a	Standard Tolerance Limits		Clear- ance ^a	Standard Tolerance Limits		Clear- ance ^a	Standard Tolerance Limits		Clear- ance ^a	Standard Tolerance Limits	
			Hole H8	Shaft e7		Hole H9	Shaft e8		Hole H9	Shaft d8		Hole H10	Shaft c9		Hole H11	Shaft
Over	To	Values shown below are in thousandths of an inch														
0 – 0.12	0.6	+0.6	– 0.6	0.6	+1.0	– 0.6	1.0	+1.0	– 1.0	2.5	+1.6	– 2.5	4.0	+2.5	– 4.0	
	1.6	0	– 1.0	2.2	0	– 1.2	2.6	0	– 1.6	5.1	0	– 3.5	8.1	0	– 5.6	
0.12 – 0.24	0.8	+0.7	– 0.8	0.8	+1.2	– 0.8	1.2	+1.2	– 1.2	2.8	+1.8	– 2.8	4.5	+3.0	– 4.5	
	2.0	0	– 1.3	2.7	0	– 1.5	3.1	0	– 1.9	5.8	0	– 4.0	9.0	0	– 6.0	
0.24 – 0.40	1.0	+0.9	– 1.0	1.0	+1.4	– 1.0	1.6	+1.4	– 1.6	3.0	+2.2	– 3.0	5.0	+3.5	– 5.0	
	2.5	0	– 1.6	3.3	0	– 1.9	3.9	0	– 2.5	6.6	0	– 4.4	10.7	0	– 7.2	
0.40 – 0.71	1.2	+1.0	– 1.2	1.2	+1.6	– 1.2	2.0	+1.6	– 2.0	3.5	+2.8	– 3.5	6.0	+4.0	– 6.0	
	2.9	0	– 1.9	3.8	0	– 2.2	4.6	0	– 3.0	7.9	0	– 5.1	12.8	0	– 8.8	
0.71 – 1.19	1.6	+1.2	– 1.6	1.6	+2.0	– 1.6	2.5	+2.0	– 2.5	4.5	+3.5	– 4.5	7.0	+5.0	– 7.0	
	3.6	0	– 2.4	4.8	0	– 2.8	5.7	0	– 3.7	10.0	0	– 6.5	15.5	0	– 10.5	
1.19 – 1.97	2.0	+1.6	– 2.0	2.0	+2.5	– 2.0	3.0	+2.5	– 3.0	5.0	+4.0	– 5.0	8.0	+6.0	– 8.0	
	4.6	0	– 3.0	6.1	0	– 3.6	7.1	0	– 4.6	11.5	0	– 7.5	18.0	0	– 12.0	
1.97 – 3.15	2.5	+1.8	– 2.5	2.5	+3.0	– 2.5	4.0	+3.0	– 4.0	6.0	+4.5	– 6.0	9.0	+7.0	– 9.0	
	5.5	0	– 3.7	7.3	0	– 4.3	8.8	0	– 5.8	13.5	0	– 9.0	20.5	0	– 13.5	
3.15 – 4.73	3.0	+2.2	– 3.0	3.0	+3.5	– 3.0	5.0	+3.5	– 5.0	7.0	+5.0	– 7.0	10.0	+9.0	– 10.0	
	6.6	0	– 4.4	8.7	0	– 5.2	10.7	0	– 7.2	15.5	0	– 10.5	24.0	0	– 15.0	
4.73 – 7.09	3.5	+2.5	– 3.5	3.5	+4.0	– 3.5	6.0	+4.0	– 6.0	8.0	+6.0	– 8.0	12.0	+10.0	– 12.0	
	7.6	0	– 5.1	10.0	0	– 6.0	12.5	0	– 8.5	18.0	0	– 12.0	28.0	0	– 18.0	
7.09 – 9.85	4.0	+2.8	– 4.0	4.0	+4.5	– 4.0	7.0	+4.5	– 7.0	10.0	+7.0	– 10.0	15.0	+12.0	– 15.0	
	8.6	0	– 5.8	11.3	0	– 6.8	14.3	0	– 9.8	21.5	0	– 14.5	34.0	0	– 22.0	
9.85 – 12.41	5.0	+3.0	– 5.0	5.0	+5.0	– 5.0	8.0	+5.0	– 8.0	12.0	+8.0	– 12.0	18.0	+12.0	– 18.0	
	10.0	0	– 7.0	13.0	0	– 8.0	16.0	0	– 11.0	25.0	0	– 17.0	38.0	0	– 26.0	
12.41 – 15.75	6.0	+3.5	– 6.0	6.0	+6.0	– 6.0	10.0	+6.0	– 10.0	14.0	+9.0	– 14.0	22.0	+14.0	– 22.0	
	11.7	0	– 8.2	15.5	0	– 9.5	19.5	0	– 13.5	29.0	0	– 20.0	45.0	0	– 31.0	
15.75 – 19.69	8.0	+4.0	– 8.0	8.0	+6.0	– 8.0	12.0	+6.0	– 12.0	16.0	+10.0	– 16.0	25.0	+16.0	– 25.0	
	14.5	0	– 10.5	18.0	0	– 12.0	22.0	0	– 16.0	32.0	0	– 22.0	51.0	0	– 35.0	

Interference Fits:

- Interference fits are those in which the inside member is larger than the outside member, requiring the application of force during assembly.
- There is some deformation of the parts after assembly, and pressure exists at the mating surfaces.
- A similar fit is obtained by shrink fitting, in which one member is heated to expand it while the other remains cool
- The interference fit creates pressure acting at the mating surfaces. The pressure causes stress in each part. Under heavy force fits, or even lighter fits in fragile parts, the stresses developed can be great enough to yield ductile materials.
- Within this standard, there are 5 classes, FN1 through FN5:
 - FN1 (light drive fit): Only light pressure required to assemble mating parts. Used for fragile parts and where no large forces must be transmitted across the joint.
 - FN2 (medium drive fit): General purpose class used frequently for steel parts of moderate cross section.
 - FN3 (heavy drive fit): Used for heavy steel parts.
 - FN4 (force fit): Used for high-strength assemblies where high resulting pressures are required.
 - FN5 (force fit): Similar to FN4 for higher pressures.

ANSI Standard Force and Shrink Fits ANSI B4.1-1967 (R1987)

Nominal Size Range, Inches		Class FN 1			Class FN 2			Class FN 3			Class FN 4			Class FN 5		
		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits	
			Hole H6	Shaft		Hole H7	Shaft s6		Hole H7	Shaft t6		Hole H7	Shaft u6		Hole H8	Shaft x7
Over	To	Values shown below are in thousandths of an inch														
0–	0.12	0.05 0.5	+0.25 0	+0.5 +0.3	0.2 0.85	+0.4 0	+0.85 +0.6				0.3 0.95	+0.4 0	+0.95 +0.7	0.3 1.3	+0.6 0	+1.3 +0.9
0.12–	0.24	0.1 0.6	+0.3 0	+0.6 +0.4	0.2 1.0	+0.5 0	+1.0 +0.7				0.4 1.2	+0.5 0	+1.2 +0.9	0.5 1.7	+0.7 0	+1.7 +1.2
0.24–	0.40	0.1 0.75	+0.4 0	+0.75 +0.5	0.4 1.4	+0.6 0	+1.4 +1.0				0.6 1.6	+0.6 0	+1.6 +1.2	0.5 2.0	+0.9 0	+2.0 +1.4
0.40–	0.56	0.1 0.8	+0.4 0	+0.8 +0.5	0.5 1.6	+0.7 0	+1.6 +1.2				0.7 1.8	+0.7 0	+1.8 +1.4	0.6 2.3	+1.0 0	+2.3 +1.6
0.56–	0.71	0.2 0.9	+0.4 0	+0.9 +0.6	0.5 1.6	+0.7 0	+1.6 +1.2				0.7 1.8	+0.7 0	+1.8 +1.4	0.8 2.5	+1.0 0	+2.5 +1.8
0.71–	0.95	0.2 1.1	+0.5 0	+1.1 +0.7	0.6 1.9	+0.8 0	+1.9 +1.4				0.8 2.1	+0.8 0	+2.1 +1.6	1.0 3.0	+1.2 0	+3.0 +2.2
0.95–	1.19	0.3 1.2	+0.5 0	+1.2 +0.8	0.6 1.9	+0.8 0	+1.9 +1.4	0.8 2.1	+0.8 0	+2.1 +1.6	+1.0 2.3	+0.8 0	+2.3 +1.8	1.3 3.3	+1.2 0	+3.3 +2.5
1.19–	1.58	0.3 1.3	+0.6 0	+1.3 +0.9	0.8 2.4	+1.0 0	+2.4 +1.8	1.0 2.6	+1.0 0	+2.6 +2.0	1.5 3.1	+1.0 0	+3.1 +2.5	1.4 4.0	+1.6 0	+4.0 +3.0
1.58–	1.97	0.4 1.4	+0.6 0	+1.4 +1.0	0.8 2.4	+1.0 0	+2.4 +1.8	1.2 2.8	+1.0 0	+2.8 +2.2	1.8 3.4	+1.0 0	+3.4 +2.8	2.4 5.0	+1.6 0	+5.0 +4.0
1.97–	2.56	0.6 1.8	+0.7 0	+1.8 +1.3	0.8 2.7	+1.2 0	+2.7 +2.0	1.3 3.2	+1.2 0	+3.2 +2.5	2.3 4.2	+1.2 0	+4.2 +3.5	3.2 6.2	+1.8 0	+6.2 +5.0
2.56–	3.15	0.7 1.9	+0.7 0	+1.9 +1.4	1.0 2.9	+1.2 0	+2.9 +2.2	1.8 3.7	+1.2 0	+3.7 +3.0	2.8 4.7	+1.2 0	+4.7 +4.0	4.2 7.2	+1.8 0	+7.2 +6.0
3.15–	3.94	0.9 2.4	+0.9 0	+2.4 +1.8	1.4 3.7	+1.4 0	+3.7 +2.8	2.1 4.4	+1.4 0	+4.4 +3.5	3.6 5.9	+1.4 0	+5.9 +5.0	4.8 8.4	+2.2 0	+8.4 +7.0
3.94–	4.73	1.1 2.6	+0.9 0	+2.6 +2.0	1.6 3.9	+1.4 0	+3.9 +3.0	2.6 4.9	+1.4 0	+4.9 +4.0	4.6 6.9	+1.4 0	+6.9 +6.0	5.8 9.4	+2.2 0	+9.4 +8.0

ANSI Standard Force and Shrink Fits ANSI B4.1-1967 (R1987)

Nominal Size Range, Inches Over To		Class FN 1			Class FN 2			Class FN 3			Class FN 4			Class FN 5		
		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits	
			Hole H6	Shaft		Hole H7	Shaft s6		Hole H7	Shaft t6		Hole H7	Shaft u6		Hole H8	Shaft x7
Values shown below are in thousandths of an inch																
4.73– 5.52	1.2	+1.0	+2.9	1.9	+1.6	+4.5	3.4	+1.6	+6.0	5.4	+1.6	+8.0	7.5	+2.5	+11.6	
	2.9	0	+2.2	4.5	0	+3.5	6.0	0	+5.0	8.0	0	+7.0	11.6	0	+10.0	
5.52– 6.30	1.5	+1.0	+3.2	2.4	+1.6	+5.0	3.4	+1.6	+6.0	5.4	+1.6	+8.0	9.5	+2.5	+13.6	
	3.2	0	+2.5	5.0	0	+4.0	6.0	0	+5.0	8.0	0	+7.0	13.6	0	+12.0	
6.30– 7.09	1.8	+1.0	+3.5	2.9	+1.6	+5.5	4.4	+1.6	+7.0	6.4	+1.6	+9.0	9.5	+2.5	+13.6	
	3.5	0	+2.8	5.5	0	+4.5	7.0	0	+6.0	9.0	0	+8.0	13.6	0	+12.0	
7.09– 7.88	1.8	+1.2	+3.8	3.2	+1.8	+6.2	5.2	+1.8	+8.2	7.2	+1.8	+10.2	11.2	+2.8	+15.8	
	3.8	0	+3.0	6.2	0	+5.0	8.2	0	+7.0	10.2	0	+9.0	15.8	0	+14.0	
7.88– 8.86	2.3	+1.2	+4.3	3.2	+1.8	+6.2	5.2	+1.8	+8.2	8.2	+1.8	+11.2	13.2	+2.8	+17.8	
	4.3	0	+3.5	6.2	0	+5.0	8.2	0	+7.0	11.2	0	+10.0	17.8	0	+16.0	
8.86– 9.85	2.3	+1.2	+4.3	4.2	+1.8	+7.2	6.2	+1.8	+9.2	10.2	+1.8	+13.2	13.2	+2.8	+17.8	
	4.3	0	+3.5	7.2	0	+6.0	9.2	0	+8.0	13.2	0	+12.0	17.8	0	+16.0	
9.85– 11.03	2.8	+1.2	+4.9	4.0	+2.0	+7.2	7.0	+2.0	+10.2	10.0	+2.0	+13.2	15.0	+3.0	+20.0	
	4.9	0	+4.0	7.2	0	+6.0	10.2	0	+9.0	13.2	0	+12.0	20.0	0	+18.0	
11.03– 12.41	2.8	+1.2	+4.9	5.0	+2.0	+8.2	7.0	+2.0	+10.2	12.0	+2.0	+15.2	17.0	+3.0	+22.0	
	4.9	0	+4.0	8.2	0	+7.0	10.2	0	+9.0	15.2	0	+14.0	22.0	0	+20.0	
12.41– 13.98	3.1	+1.4	+5.5	5.8	+2.2	+9.4	7.8	+2.2	+11.4	13.8	+2.2	+17.4	18.5	+3.5	+24.2	
	5.5	0	+4.5	9.4	0	+8.0	11.4	0	+10.0	17.4	0	+16.0	24.2	0	+22.0	
13.98– 15.75	3.6	+1.4	+6.1	5.8	+2.2	+9.4	9.8	+2.2	+13.4	15.8	+2.2	+19.4	21.5	+3.5	+27.2	
	6.1	0	+5.0	9.4	0	+8.0	13.4	0	+12.0	19.4	0	+18.0	27.2	0	+25.0	
15.75– 17.72	4.4	+1.6	+7.0	6.5	+2.5	+10.6	+9.5	+2.5	+13.6	17.5	+2.5	+21.6	24.0	+4.0	+30.5	
	7.0	0	+6.0	10.6	0	+9.0	13.6	0	+12.0	21.6	0	+20.0	30.5	0	+28.0	
17.72– 19.69	4.4	+1.6	+7.0	7.5	+2.5	+11.6	11.5	+2.5	+15.6	19.5	+2.5	+23.6	26.0	+4.0	+32.5	
	7.0	0	+6.0	11.6	0	+10.0	15.6	0	+14.0	23.6	0	+22.0	32.5	0	+30.0	

Locational Fits (LC, LT, and LN):

Locational fits are fits intended to determine only the location of the mating parts; they may provide rigid or accurate location, as with interference fits, or provide some freedom of location, as with clearance fits. Accordingly, they are divided into three groups: clearance fits (LC), transition fits (LT), and interference fits (LN).

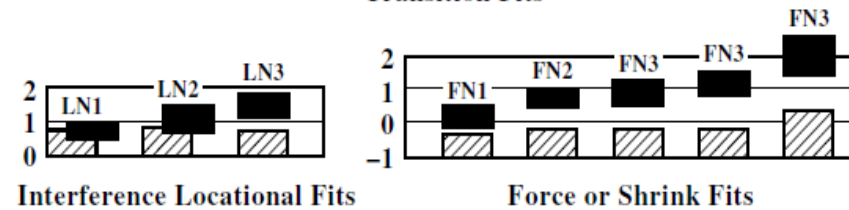
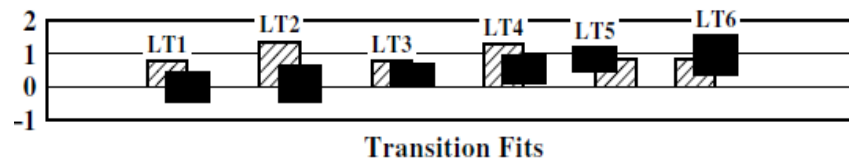
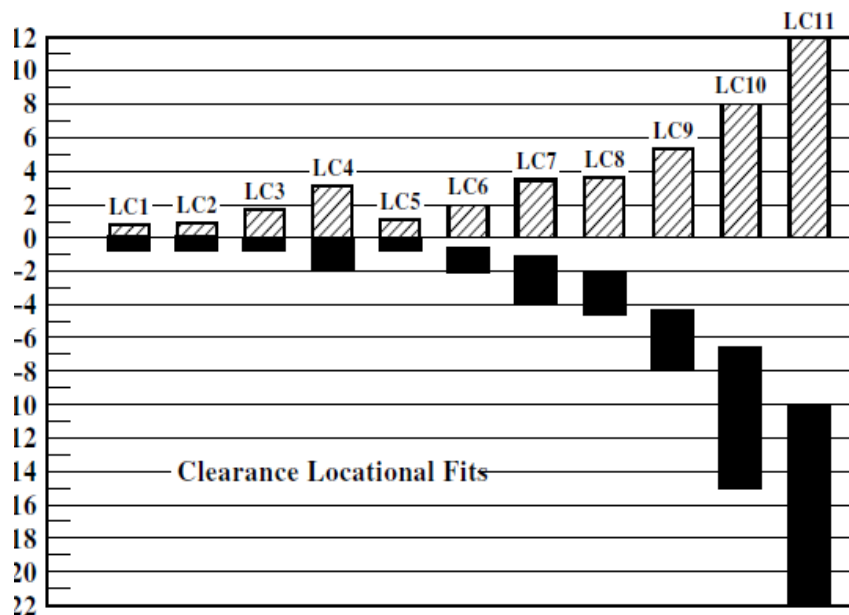
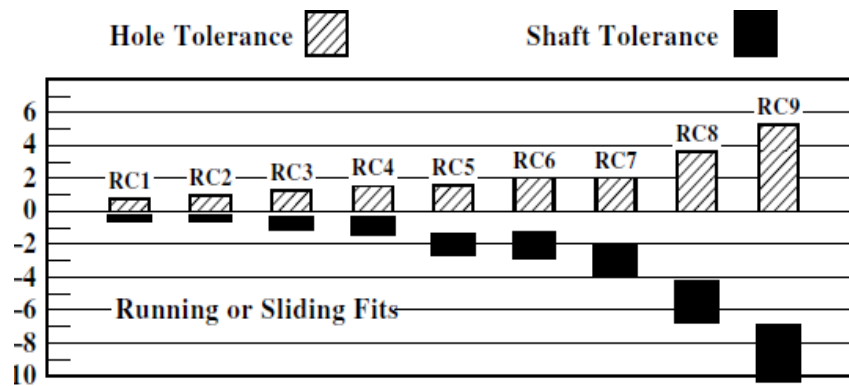
These are described as follows:

LC Locational clearance fits: are intended for parts which are normally stationary, but that can be freely assembled or disassembled. They range from snug fits for parts requiring accuracy of location, through the medium clearance fits for parts such as spigots, to the looser fastener fits where freedom of assembly is of prime importance.

LT Locational transition fits: are a compromise between clearance and interference fits, for applications where accuracy of location is important, but either a small amount of clearance or interference is permissible. (dowel pin application)

LN Locational interference fits are used where accuracy of location is of prime importance, and for parts requiring rigidity and alignment with no special requirements for bore pressure. Such fits are not intended for parts designed to transmit frictional loads from one part to another by virtue of the tightness of fit. These conditions are covered by force fits.

Diagrams show disposition of hole and shaft tolerances (in thousandths of an inch) with respect to a diameter of 1 inch.



American National Standard Preferred Metric Limits and Fits.

ANSI B4.2-1978 (R1994) describes the ISO system of metric limits and fits for mating parts as approved for general engineering usage in the United States.

	ISO SYMBOL		DESCRIPTION	
	Hole Basis	Shaft Basis		
Clearance Fits	H11/c11	C11/h11	<u>Loose running</u> fit for wide commercial tolerances or allowances on external members.	↑ More Clearance
	H9/d9	D9/h9	<u>Free running</u> fit not for use where accuracy is essential, but good for large temperature variations, high running speeds, or heavy journal pressures.	
	H8/f7	F8/h7	<u>Close Running</u> fit for running on accurate machines and for accurate moderate speeds and journal pressures.	
	H7/g6	G7/h6	<u>Sliding fit</u> not intended to run freely, but to move and turn freely and locate accurately.	
	H7/h6	H7/h6	<u>Locational clearance</u> fit provides snug fit for locating stationary parts; but can be freely assembled and disassembled.	
Transition Fits	H7/k6	K7/h6	<u>Locational transition</u> fit for accurate location, a compromise between clearance and interference.	↓ More Interference
	H7/n6	N7/h6	<u>Locational transition</u> fit for more accurate location where greater interference is permissible.	
Interference Fits	H7/p6 ^a	P7/h6	<u>Locational interference</u> fit for parts requiring rigidity and alignment with prime accuracy of location but without special bore pressure requirements.	
	H7/s6	S7/h6	<u>Medium drive</u> fit for ordinary steel parts or shrink fits on light sections, the tightest fit usable with cast iron.	
	H7/u6	U7/h6	<u>Force</u> fit suitable for parts which can be highly stressed or for shrink fits where the heavy pressing forces required are impractical.	

Basic hole system; Allowance based on the hole at MMC

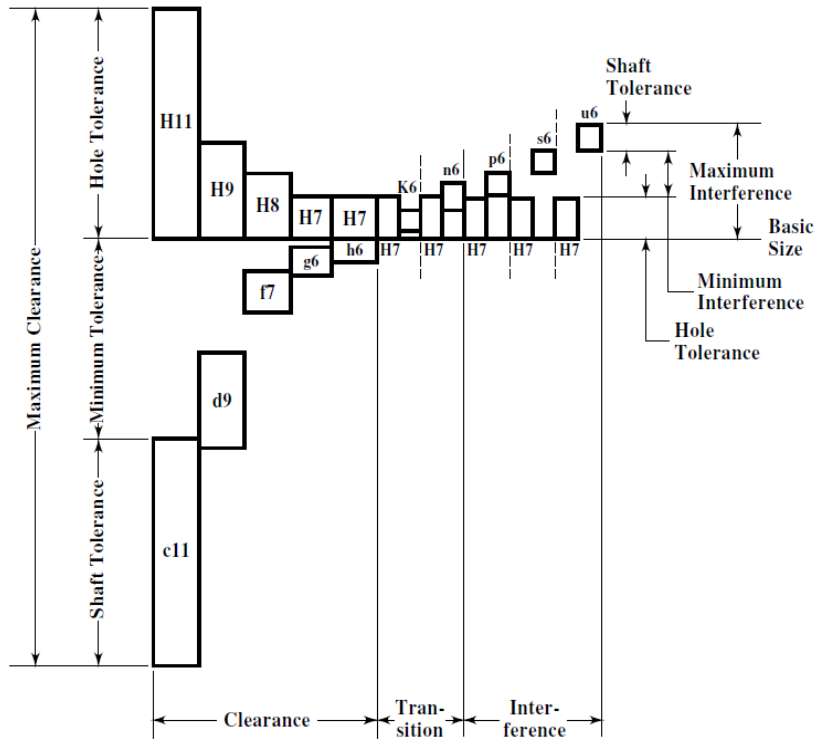
- The basic size is always equal to the *minimum hole size (Hole MMC)*
- *Subtract the allowance* from the basic size to get the *maximum shaft size (Shaft MMC)*
- Apply the appropriate tolerances to the hole and to the shaft (Hole & Shaft LMC)

Basic shaft system; Allowance base on the shaft at MMC

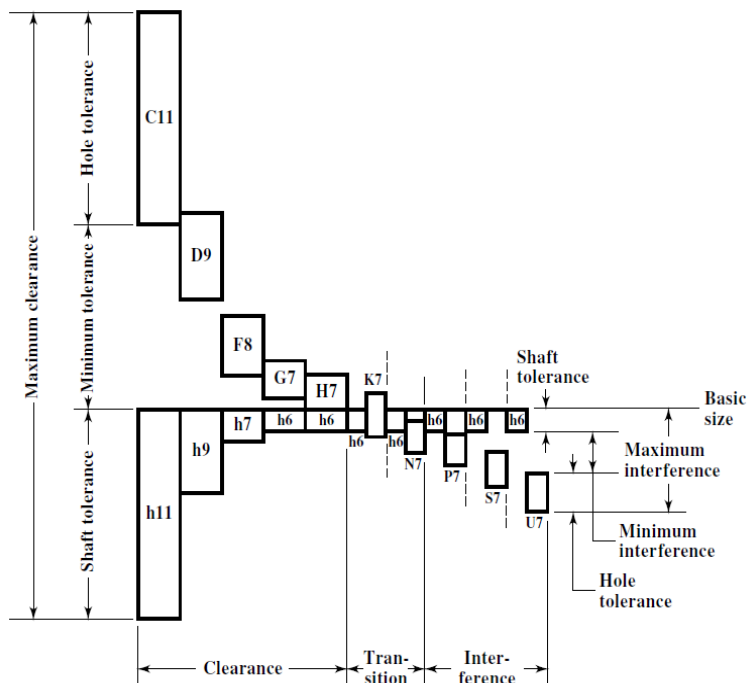
- Basic size is always equal to the *maximum shaft size*
- *Add the allowance* to the basic size to get the *minimum hole size*
- Apply the appropriate tolerances to the shaft and the hole

Normally, the hole basis system is preferred; however, when a common shaft mates with several holes, the shaft basis system should be used

Preferred Hole Basis Fits:



Preferred Shaft Basis Fits:



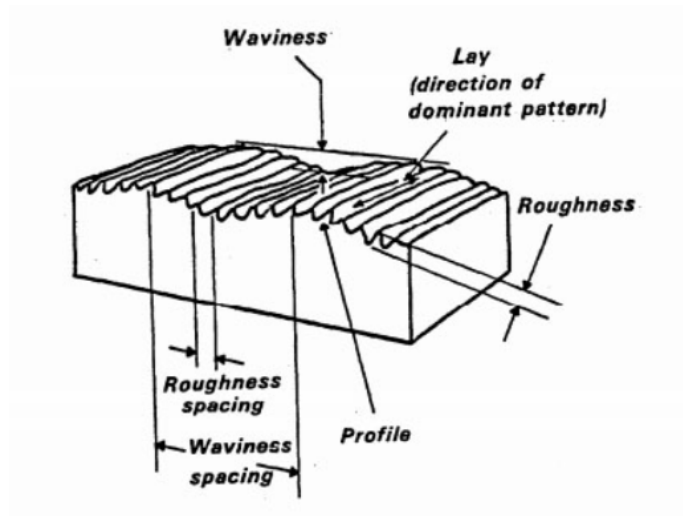
Example of the Description of Preferred Fits table:

Table 12. American National Standard Preferred Hole Basis Metric Clearance Fits ANSI B4.2-1978 (R1994)

Basic Size ^a		Loose Running			Free Running			Close Running			Sliding			Locational Clearance		
		Hole H11	Shaft C11	Fit ^b	Hole H9	Shaft d9	Fit ^b	Hole H8	Shaft f7	Fit ^b	Hole H7	Shaft g6	Fit ^b	Hole H7	Shaft h6	Fit ^b
1	Max	1.060	0.940	0.180	1.025	0.980	0.070	1.014	0.994	0.030	1.010	0.998	0.018	1.010	1.000	0.016
	Min	1.000	0.880	0.060	1.000	0.995	0.020	1.000	0.984	0.006	1.000	0.992	0.002	1.000	0.994	0.000
1.2	Max	1.260	1.140	0.180	1.225	1.180	0.070	1.214	1.194	0.030	1.210	1.198	0.018	1.210	1.200	0.016
	Min	1.200	1.080	0.060	1.200	1.155	0.020	1.200	1.184	0.006	1.200	1.192	0.002	1.200	1.194	0.000
1.6	Max	1.660	1.540	0.180	1.625	1.580	0.070	1.614	1.594	0.030	1.610	1.598	0.018	1.610	1.600	0.016
	Min	1.600	1.480	0.060	1.600	1.555	0.020	1.600	1.584	0.006	1.600	1.592	0.002	1.600	1.594	0.000
2	Max	2.060	1.940	0.180	2.025	1.980	0.070	2.014	1.994	0.030	2.010	1.998	0.018	2.010	2.000	0.016
	Min	2.000	1.880	0.060	2.000	1.955	0.020	2.000	1.984	0.006	2.000	1.992	0.002	2.000	1.994	0.000
2.5	Max	2.560	2.440	0.180	2.525	2.480	0.070	2.514	2.494	0.030	2.510	2.498	0.018	2.510	2.500	0.016
	Min	2.500	2.380	0.060	2.500	2.455	0.020	2.500	2.484	0.006	2.500	2.492	0.002	2.500	2.494	0.000
3	Max	3.060	2.940	0.180	3.025	2.980	0.070	3.014	2.994	0.030	3.010	2.998	0.018	3.010	3.000	0.016
	Min	3.000	2.880	0.060	3.000	2.955	0.020	3.000	2.984	0.006	3.000	2.992	0.002	3.000	2.994	0.000
4	Max	4.075	3.930	0.220	4.030	3.970	0.090	4.018	3.990	0.040	4.012	3.996	0.024	4.012	4.000	0.020
	Min	4.000	3.855	0.070	4.000	3.940	0.030	4.000	3.978	0.010	4.000	3.988	0.004	4.000	3.992	0.000
5	Max	5.075	4.930	0.220	5.030	4.970	0.090	5.018	4.990	0.040	5.012	4.996	0.024	5.012	5.000	0.020
	Min	5.000	4.855	0.070	5.000	4.940	0.030	5.000	4.978	0.010	5.000	4.988	0.004	5.000	4.992	0.000
6	Max	6.075	5.930	0.220	6.030	5.970	0.090	6.018	5.990	0.040	6.012	5.996	0.024	6.012	6.000	0.020
	Min	6.000	5.855	0.070	6.000	5.940	0.030	6.000	5.978	0.010	6.000	5.988	0.004	6.000	5.992	0.000
8	Max	8.090	7.920	0.260	8.036	7.960	0.112	8.022	7.987	0.050	8.015	7.995	0.029	8.015	8.000	0.024
	Min	8.000	7.830	0.080	8.000	7.924	0.040	8.000	7.972	0.013	8.000	7.986	0.005	8.000	7.991	0.000
10	Max	10.090	9.920	0.260	10.036	9.960	0.112	10.022	9.987	0.050	10.015	9.995	0.029	10.015	10.000	0.024
	Min	10.000	9.830	0.080	10.000	9.924	0.040	10.000	9.972	0.013	10.000	9.986	0.005	10.000	9.991	0.000
12	Max	12.110	11.905	0.315	12.043	11.956	0.136	12.027	11.984	0.061	12.018	11.994	0.035	12.018	12.000	0.029
	Min	12.000	11.795	0.095	12.000	11.907	0.050	12.000	11.966	0.016	12.000	11.983	0.006	12.000	11.989	0.000
16	Max	16.110	15.905	0.315	16.043	15.950	0.136	16.027	15.984	0.061	16.018	15.994	0.035	16.018	16.000	0.029
	Min	16.000	15.795	0.095	16.000	15.907	0.050	16.000	15.966	0.016	16.000	15.983	0.006	16.000	15.989	0.000
20	Max	20.130	19.890	0.370	20.052	19.935	0.169	20.033	19.980	0.074	20.021	19.993	0.041	20.021	20.000	0.034
	Min	20.000	19.760	0.110	20.000	19.883	0.065	20.000	19.959	0.020	20.000	19.980	0.007	20.000	19.987	0.000
25	Max	25.130	24.890	0.370	25.052	24.935	0.169	25.033	24.980	0.074	25.021	24.993	0.041	25.021	25.000	0.034
	Min	25.000	24.760	0.110	25.000	24.883	0.065	25.000	24.959	0.020	25.000	24.980	0.007	25.000	24.987	0.000

American National Standard Preferred Hole Basis Metric Clearance Fits ANSI B4.2-1978 (R1994)

Basic Size ^a		Loose Running			Free Running			Close Running			Sliding			Locational Clearance		
		Hole H11	Shaft C11	Fit ^b	Hole H9	Shaft d9	Fit ^b	Hole H8	Shaft f7	Fit ^b	Hole H7	Shaft g6	Fit ^b	Hole H7	Shaft h6	Fit ^b
30	Max	30.130	29.890	0.370	30.052	29.935	0.169	30.033	29.980	0.074	30.021	29.993	0.041	30.021	30.000	0.034
	Min	30.000	29.760	0.110	30.000	29.883	0.065	30.000	29.959	0.020	30.000	29.980	0.007	30.000	29.987	0.000
40	Max	40.160	39.880	0.440	40.062	39.920	0.204	40.039	39.975	0.089	40.025	39.991	0.050	40.025	40.000	0.041
	Min	40.000	39.720	0.120	40.000	39.858	0.080	40.000	39.950	0.025	40.000	39.975	0.009	40.000	39.984	0.000
50	Max	50.160	49.870	0.450	50.062	49.920	0.204	50.039	49.975	0.089	50.025	49.991	0.050	50.025	50.000	0.041
	Min	50.000	49.710	0.130	50.000	49.858	0.080	50.000	49.950	0.025	50.000	49.975	0.009	50.000	49.984	0.000
60	Max	60.190	59.860	0.520	60.074	59.900	0.248	60.046	59.970	0.106	60.030	59.990	0.059	60.030	60.000	0.049
	Min	60.000	59.670	0.140	60.000	59.826	0.100	60.000	59.940	0.030	60.000	59.971	0.010	60.000	59.981	0.000
80	Max	80.190	79.850	0.530	80.074	79.900	0.248	80.046	79.970	0.106	80.030	79.990	0.059	80.030	80.000	0.049
	Min	80.000	79.660	0.150	80.000	79.826	0.100	80.000	79.940	0.030	80.000	79.971	0.010	80.000	79.981	0.000
100	Max	100.220	99.830	0.610	100.087	99.880	0.294	100.054	99.964	0.125	100.035	99.988	0.069	100.035	100.000	0.057
	Min	100.000	99.610	0.170	100.000	99.793	0.120	100.000	99.929	0.036	100.000	99.966	0.012	100.000	99.978	0.000
120	Max	120.220	119.820	0.620	120.087	119.880	0.294	120.054	119.964	0.125	120.035	119.988	0.069	120.035	120.000	0.057
	Min	120.000	119.600	0.180	120.000	119.793	0.120	120.000	119.929	0.036	120.000	119.966	0.012	120.000	119.978	0.000
160	Max	160.250	159.790	0.710	160.100	159.855	0.345	160.063	159.957	0.146	160.040	159.986	0.079	160.040	160.000	0.065
	Min	160.000	159.540	0.210	160.000	159.755	0.145	160.000	159.917	0.043	160.000	159.961	0.014	160.000	159.975	0.000
200	Max	200.290	199.760	0.820	200.115	199.830	0.400	200.072	199.950	0.168	200.046	199.985	0.090	200.046	200.000	0.075
	Min	200.000	199.470	0.240	200.000	199.715	0.170	200.000	199.904	0.050	200.000	199.956	0.015	200.000	199.971	0.000
250	Max	250.290	249.720	0.860	250.115	249.830	0.400	250.072	249.950	0.168	250.046	249.985	0.090	250.046	250.000	0.075
	Min	250.000	249.430	0.280	250.000	249.715	0.170	250.000	249.904	0.050	250.000	249.956	0.015	250.000	249.971	0.000
300	Max	300.320	299.670	0.970	300.130	299.810	0.450	300.081	299.944	0.189	300.052	299.983	0.101	300.052	300.000	0.084
	Min	300.000	299.350	0.330	300.000	299.680	0.190	300.000	299.892	0.056	300.000	299.951	0.017	300.000	299.968	0.000
400	Max	400.360	399.600	1.120	400.140	399.790	0.490	400.089	399.938	0.208	400.057	399.982	0.111	400.057	400.000	0.093
	Min	400.000	399.240	0.400	400.000	399.650	0.210	400.000	399.881	0.062	400.000	399.946	0.018	400.000	399.964	0.000
500	Max	500.400	499.520	1.280	500.155	499.770	0.540	500.097	499.932	0.228	500.063	499.980	0.123	500.063	500.000	0.103
	Min	500.000	499.120	0.480	500.000	499.615	0.230	500.000	499.869	0.068	500.000	499.940	0.020	500.000	499.960	0.000

Surface Texture Characteristics

Roughness: It consists of the finer irregularities in the surface texture resulting from the production process or material condition.

Roughness Height (Roughness average: Ra): It is rated as the arithmetical average (AA) deviation expressed in micrometers (μm).

A micrometer is one millionth of a meter (0.000001 meter).

A micro-inch (μin) is one millionth of an inch (0.000001 inch).

Example:

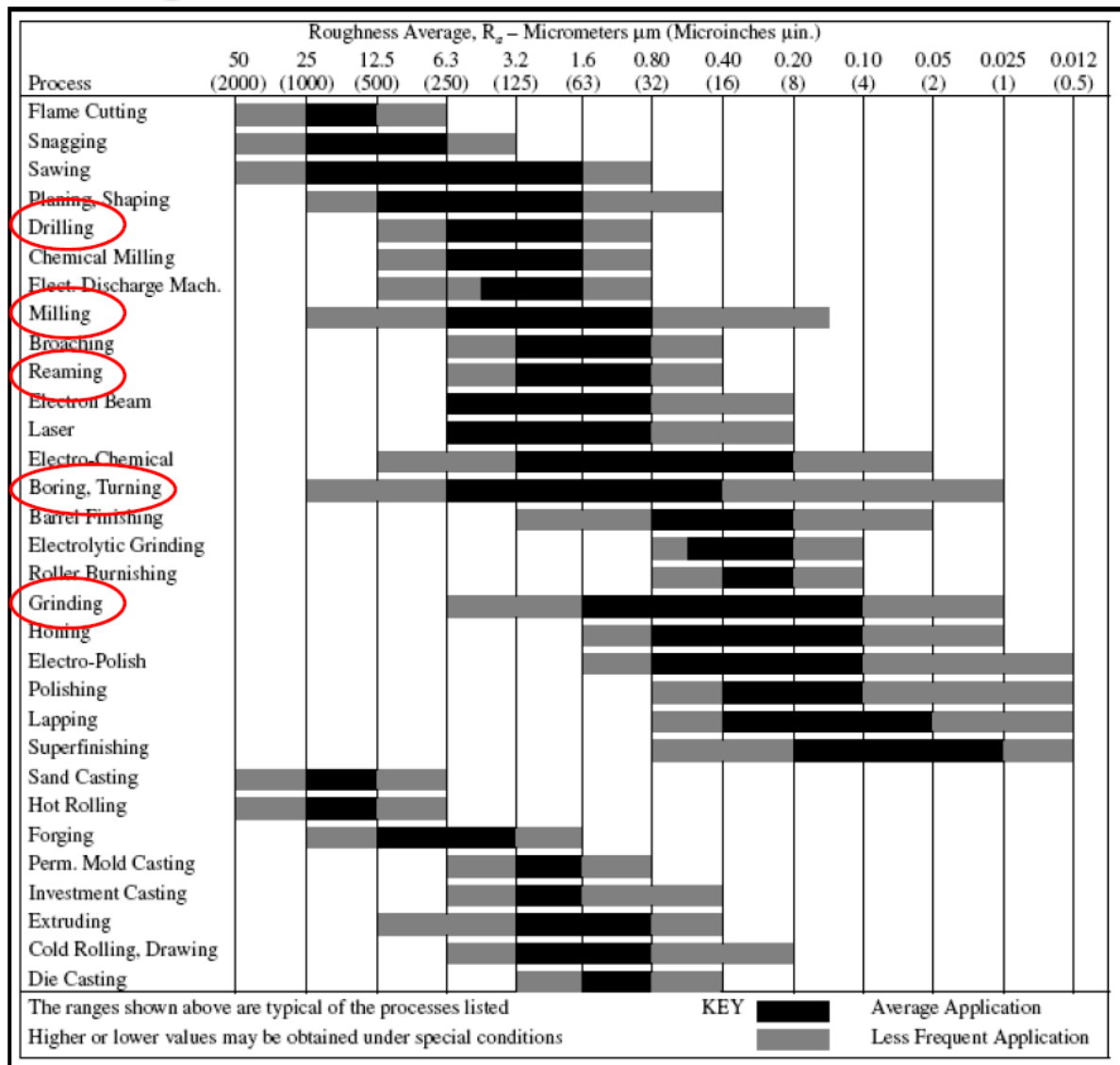
One micro-inch equals 0.0254 micrometer ($1 \mu\text{in.} = 0.0254 \mu\text{m}$).

Roughness Width: It is the distance parallel to the nominal surface between successive peaks or ridges which constitutes the predominant pattern of the roughness.

Waviness: It is the usually widely spaced component of surface texture and is generally spaced farther apart than the roughness width cut-off.

Lay: The direction of the predominant surface pattern, which is ordinarily determined by the production method used

Surface Roughness (Common Production Methods)



Surface Texture Symbol:

	Basic symbol for surface under consideration or to a specification explained elsewhere in a note. Surface may be produced by any method
	Material removal by machining is required.
3.5 	Material Removal Allowance. The number indicates the amount of stock to be removed by machining in millimeters (or inches). Tolerances may be added to the basic value shown or in general note.
	Material Removal Prohibited



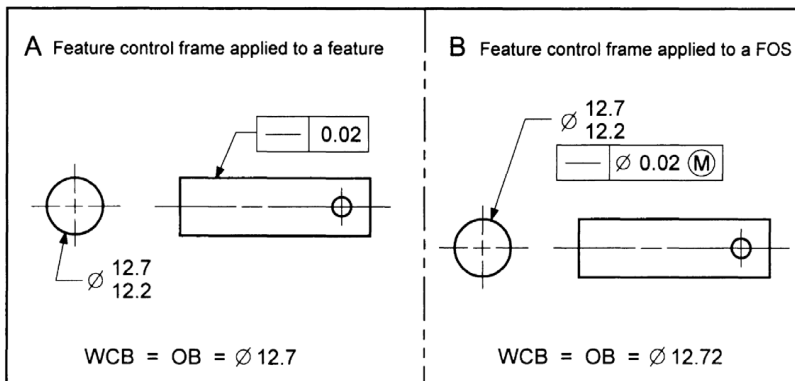
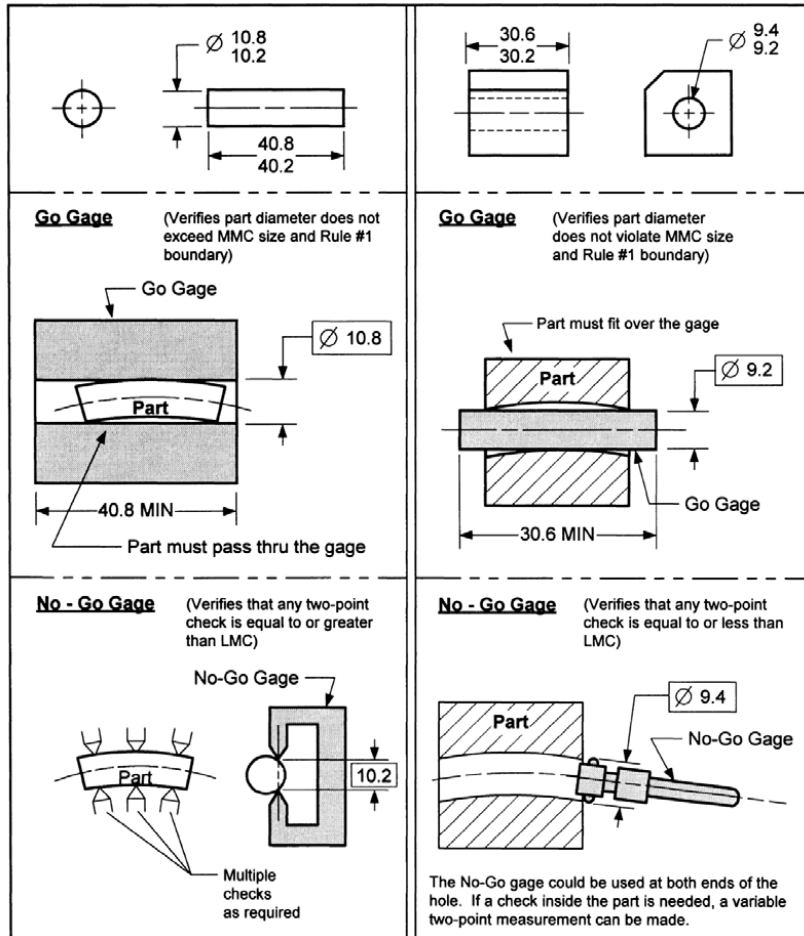
Removal of Material by Machining

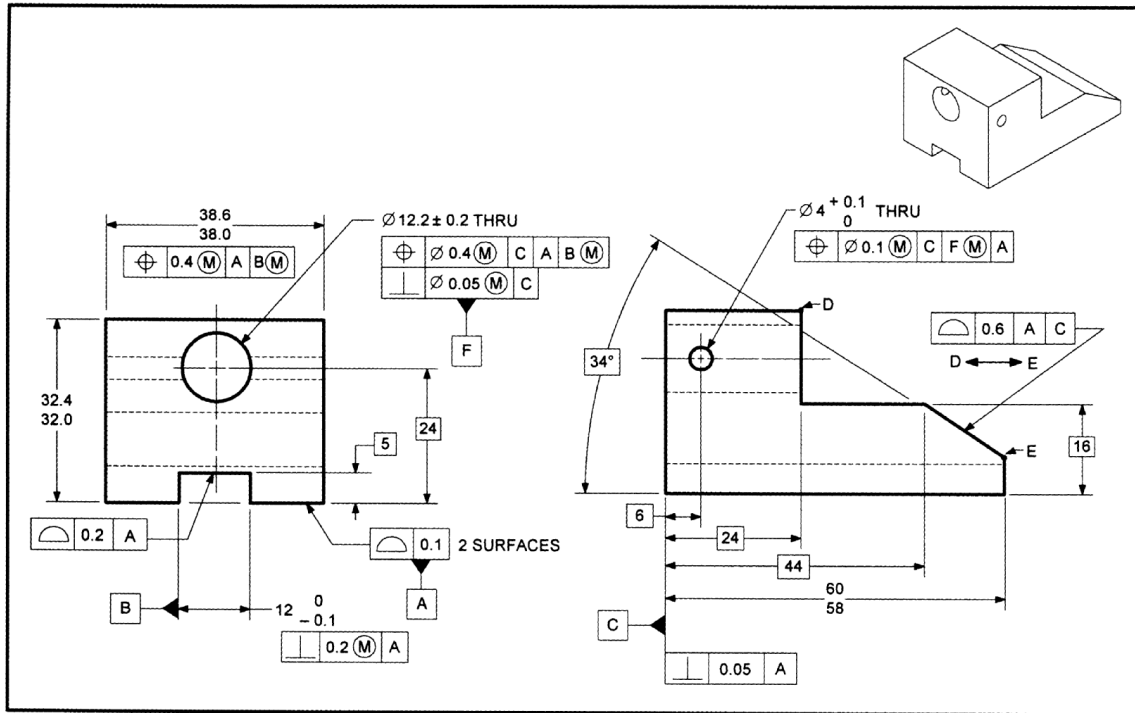
General notes on the drawing (mostly on title block)

1. All surfaces 
2. All surfaces  unless otherwise specified
3. Surfaces marked  to be 

Geometric Dimension and Tolerance GD&T:

Why GD&T:





Size dimension and actual local size:

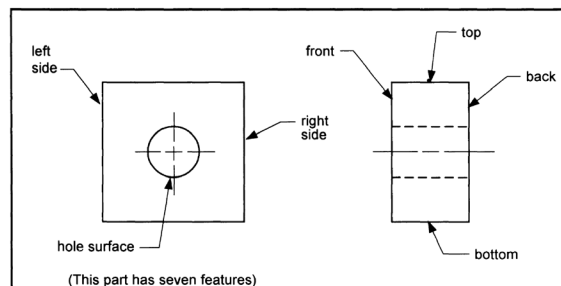
Size dimension: dimension across two opposed or partially opposed surfaces, line elements, or points

Actual local size: measured value of any individual distance at any cross section of a feature of size

Feature and complex feature:

General terms applied to a physical portion of a part such as a surface, hole or slot,

- Feature – any surface on a part
- Complex feature - Single surface of compound curvature

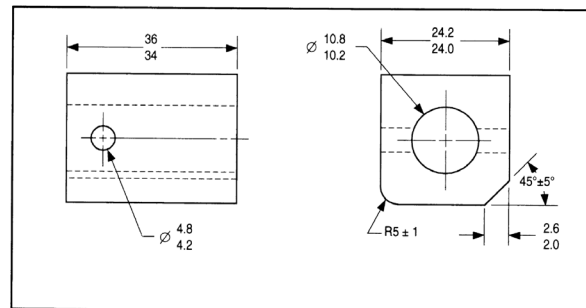


Feature of Size (FOS)

Either a cylindrical or spherical surface, or opposed parallel surfaces, associated with size dimension,

- The key to FOS features is opposing surfaces or elements
- An axis, median point or center plane can be derived from a FOS
- Is associated with a size dimension.

Non-features of size



Example of FOS & Non-FOS

Two types of Features of Size:

External FOS

- comprised of external part surfaces or elements
 - shaft diameter
 - planar length, width or height

Internal FOS

- comprised of internal part surfaces or elements
 - hole diameter
 - width of slot

Material Conditions:

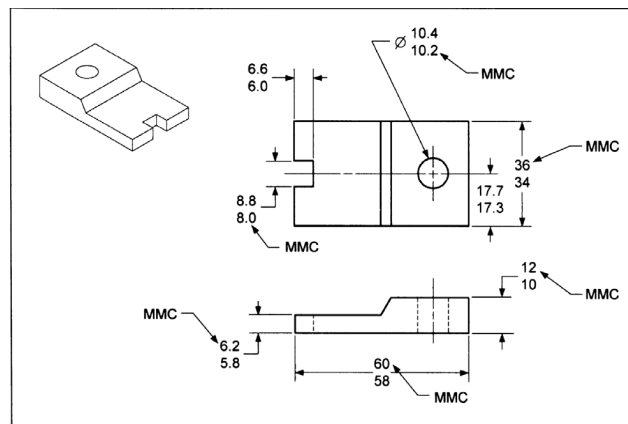
A key concept in geometric tolerancing is the ability to specify tolerances at various part feature material conditions. A geometric tolerance can be specified to apply at the largest size, smallest size, or actual size of a feature of size

Every FOS has a MMC and a LMC

a) Maximum Material Condition (MMC)

The condition in which a FOS contains the greatest amount of material allowable within the stated size limits

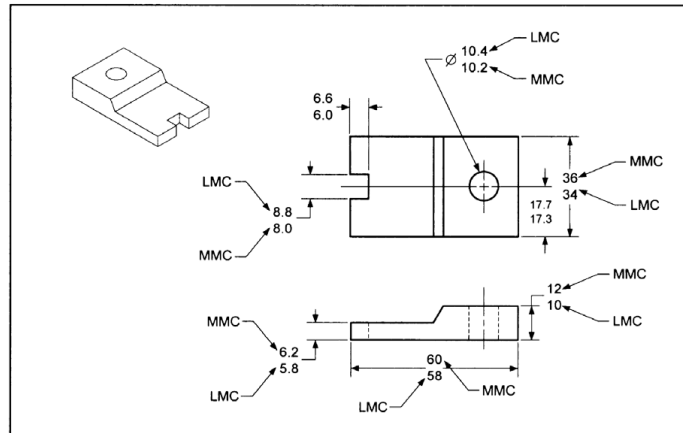
- Largest value for an external feature of size (e.g., shaft diameter)
- Smallest value for an internal feature of size (e.g., hole diameter)
- The MMC of an external FOS is the largest size limit
- The MMC of an internal FOS is the smallest size limit



b) Least Material Condition (LMC)

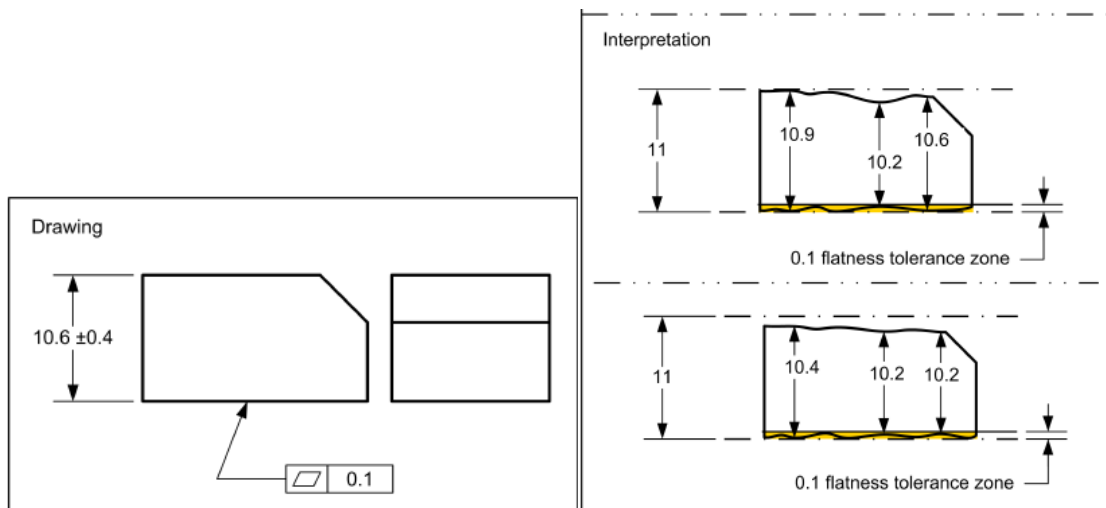
The condition in which a FOS contains the minimum amount of material allowable within the stated size limits

- Smallest value for an external feature of size
- Largest value for an internal feature of size
- The LMC of an external FOS is the smallest size limit
- The LMC of an internal FOS is the largest size limit



Regardless of Feature Size (RFS)

Is the term that indicates a geometric tolerance applies at any increment of size of the feature within its size tolerance,
e.g. a hole position may be tolerance exactly as specified regardless of the size of the hole.



Symbols and Modifiers

Geometric Characteristic Symbols

14 symbols used in the language of geometric tolerancing

Five categories: form, profile, orientation, location, and runout

Modifiers – set of symbols to communicate additional information about tolerancing of a part

Category	Characteristic	Symbol	Uses a Datum Reference
Form	Straightness		Never
	Flatness		
	Circularity (roundness)		
	Cylindricity		
Profile	Profile of a line		Sometimes
	Profile of a surface		
Location	Position		Always
	Concentricity		
	Symmetry		
Orientation	Angularity		
	Perpendicularity		
	Parallelism		
Runout	Circular runout		
	Total runout		

The Five Geometry Attributes of a Feature:

- Size – how large or small
- Location – relative to other features
- Orientation – angular relationship with a reference plane or axis
- Form – shape of a surface
- Surface texture – composite of roughness deviations typical of a real surface

Feature Control Frame

A rectangular box, divided into compartments, in which a geometric characteristic symbol, a tolerances, value, modifiers and datum references are placed.

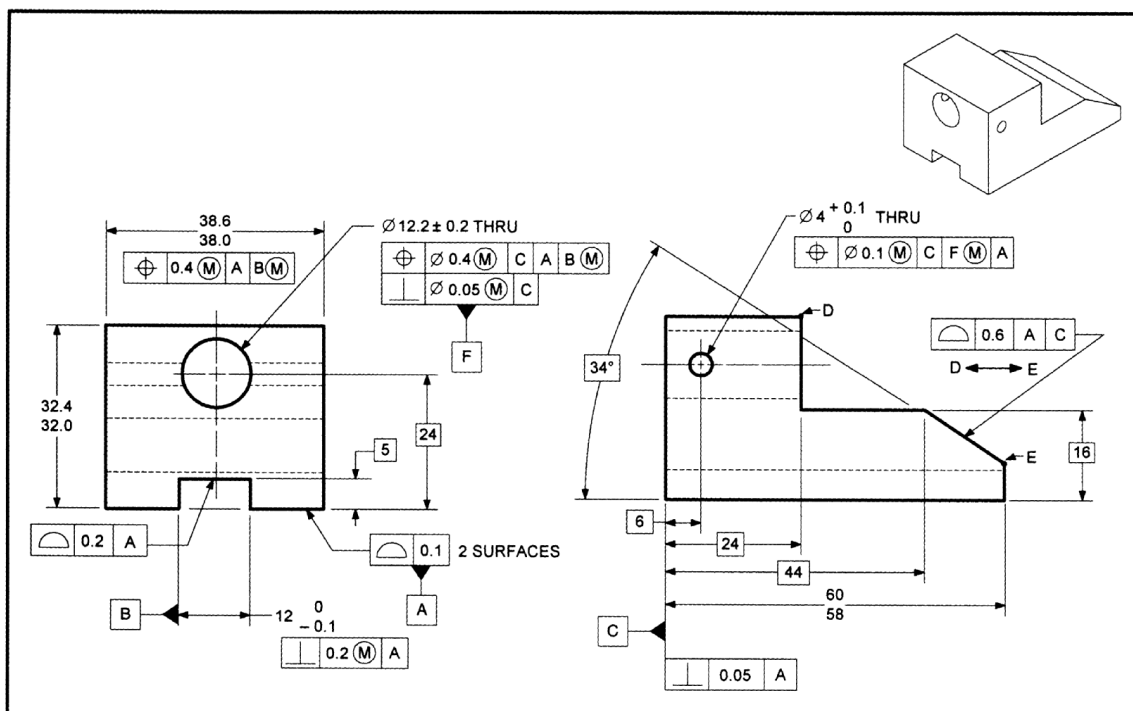
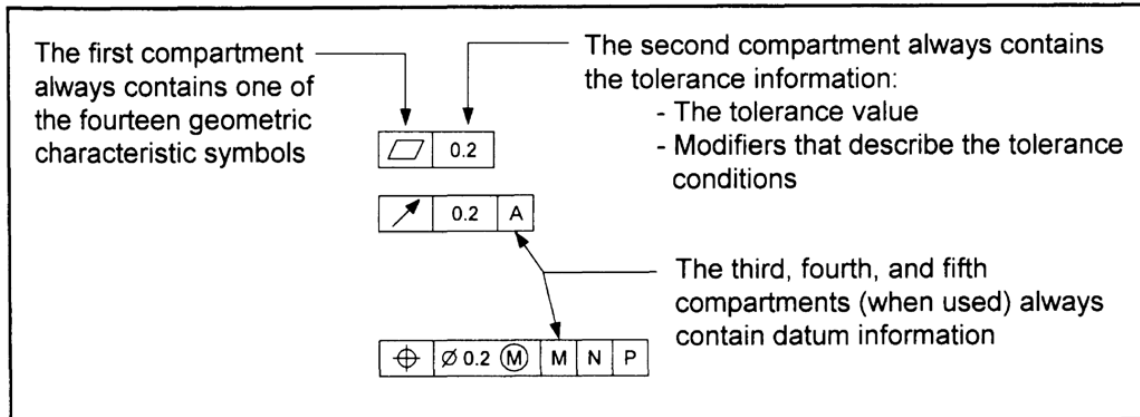
- The first compartment always contains one of the 14 geometric characteristic symbols.
- The second compartment always contains the tolerance information and modifiers that describe tolerance conditions.

MENG 3050-Power Transmission Components

- Subsequent compartments, when used, describe datum information, (datum information must be in order of primary, secondary, tertiary, etc....)

A rectangular box divided into compartments that contain:

- Geometric characteristic symbol
- Tolerance value
- Modifiers
- Datum references



The “Datum System”

Is a set of symbols and rules that communicates how dimensional measurements are to be made.

The datum system specifies which part surfaces are to come into contact with inspection equipment for measurement of a dimension.

It also specifies in which sequence the part is to come into contact with inspection equipment for measurement of a dimension.

Three Important Benefits

It aids in:

- making repeatable dimensional measurements
- communicating part functional relationships
- in making dimensional measurements as intended by the designer

Implied Datums

An implied datum is an assumed plane, axis or point from which a dimensional measurement is to be made

Shortcomings of Implied Datums

Implied datums have two major shortcomings.

They do not clearly specify which surfaces are to come into contact with inspection equipment, an assumption has to be made.

they do not indicate in which sequence the part has to come into contact with inspection equipment.

Consequences of Implied Datums

Good parts are rejected and bad parts are accepted

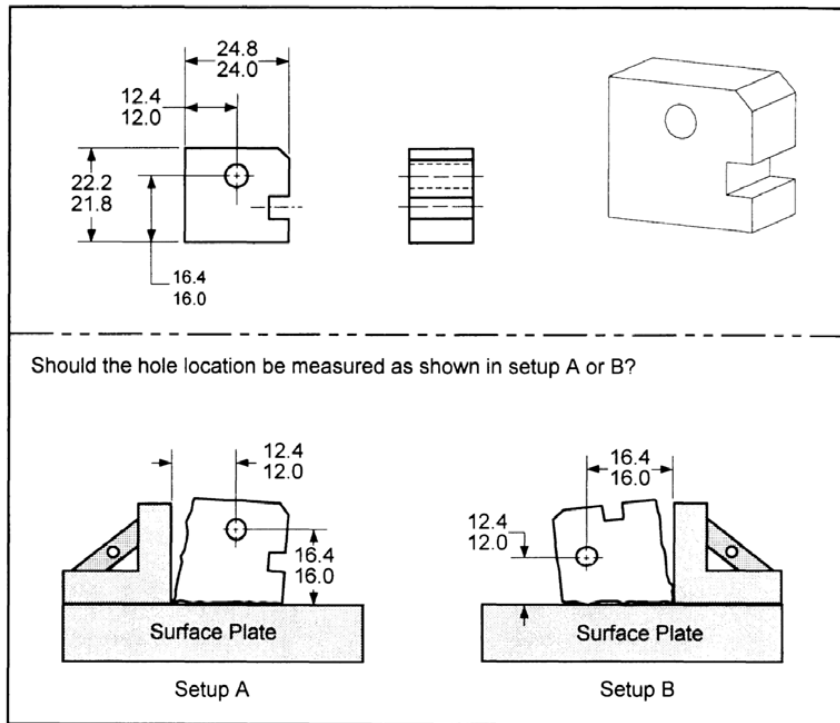
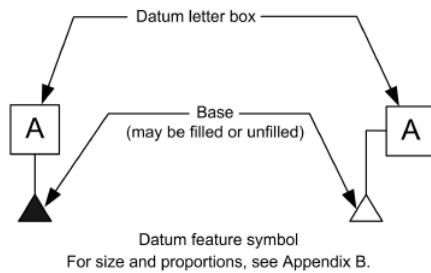


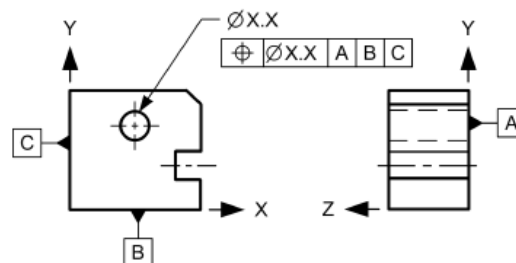
FIGURE 5-1 Implied Datums

Datum Feature Symbol

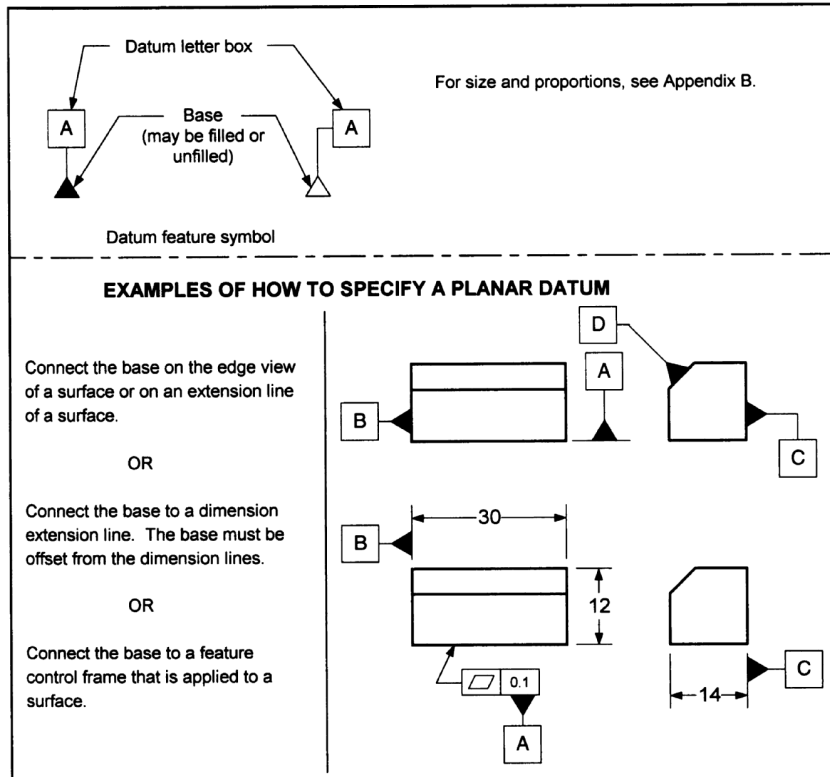
- is used to specify a datum feature on a drawing .



Datum Feature Symbol



Datum Reference Frame Symbol.

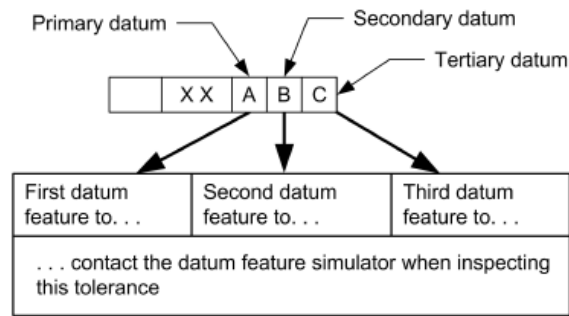


Referencing Datums in Feature Control Frames

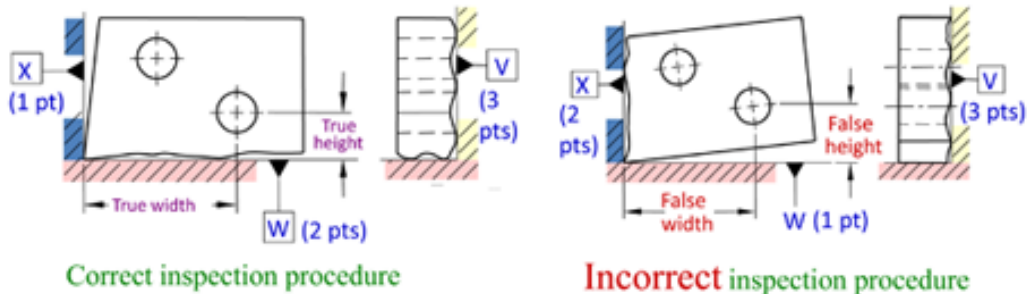
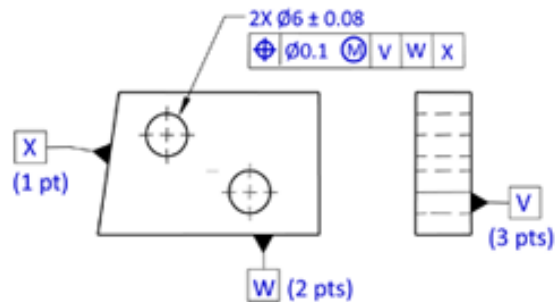
By using appropriate feature control frames, we could communicate how the datums should be used.

Feature control frames specify the sequence for contacting the part to the datums referenced. The sequence is given by reading the datums from left to right:

- The first datum to the left is the “primary datum” and must contact the inspection/assembly equipment first.
- The second datum from the left is the “secondary datum” and is the second to come into contact with the inspection equipment
- The third datum in the feature control frame is the “tertiary datum” and would contact the inspection equipment last



The order of the datums is critical!



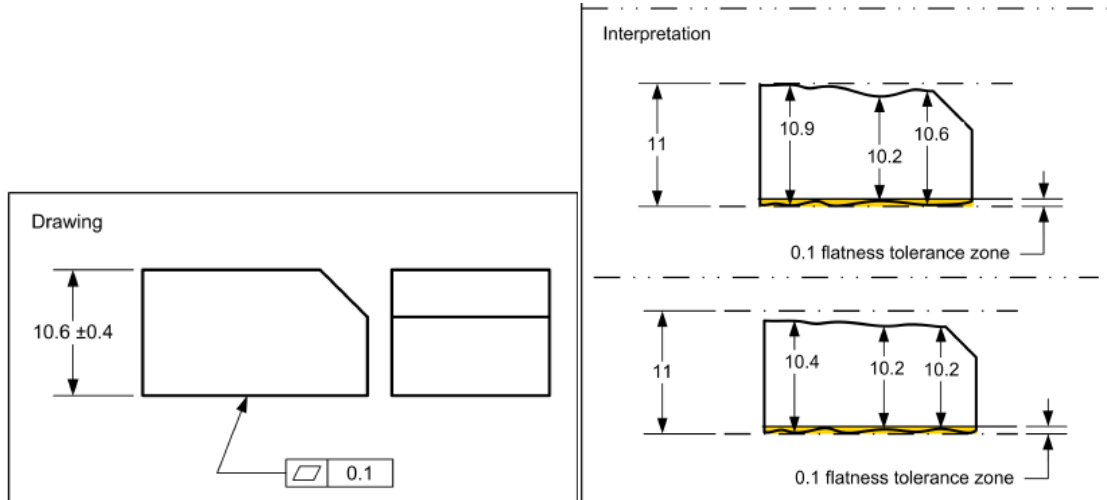
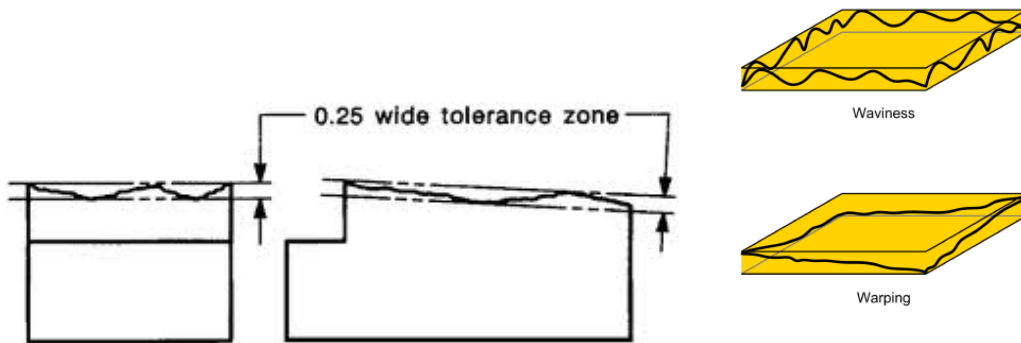
From Controls:

- Form is a characteristic that limits the shape error of a part surface (in some cases, an axis or centerplane) relative to its perfect counterpart
- Form controls limit flatness, straightness, circularity or cylindricity of part surfaces,

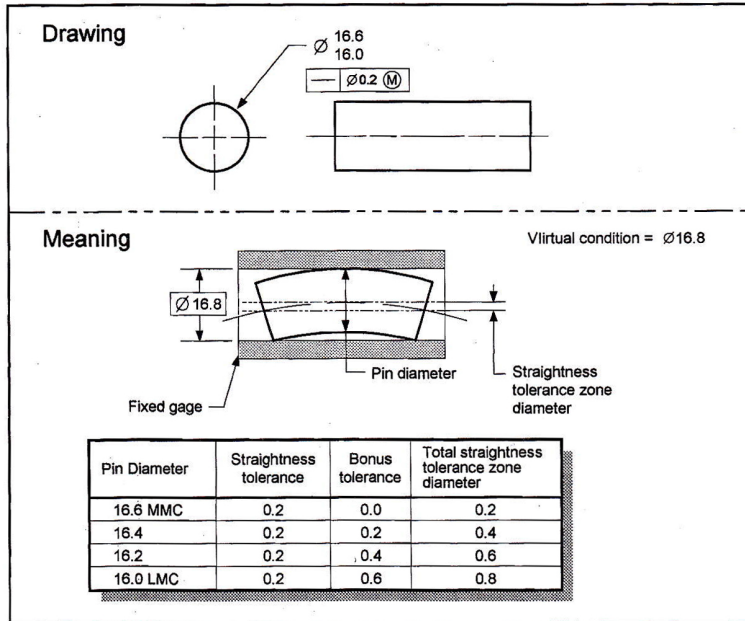
4 types of FORM geometric characteristics :

STRAIGHTNESS	—
FLATNESS	
CIRCULARITY	○
CYLINDRICITY	

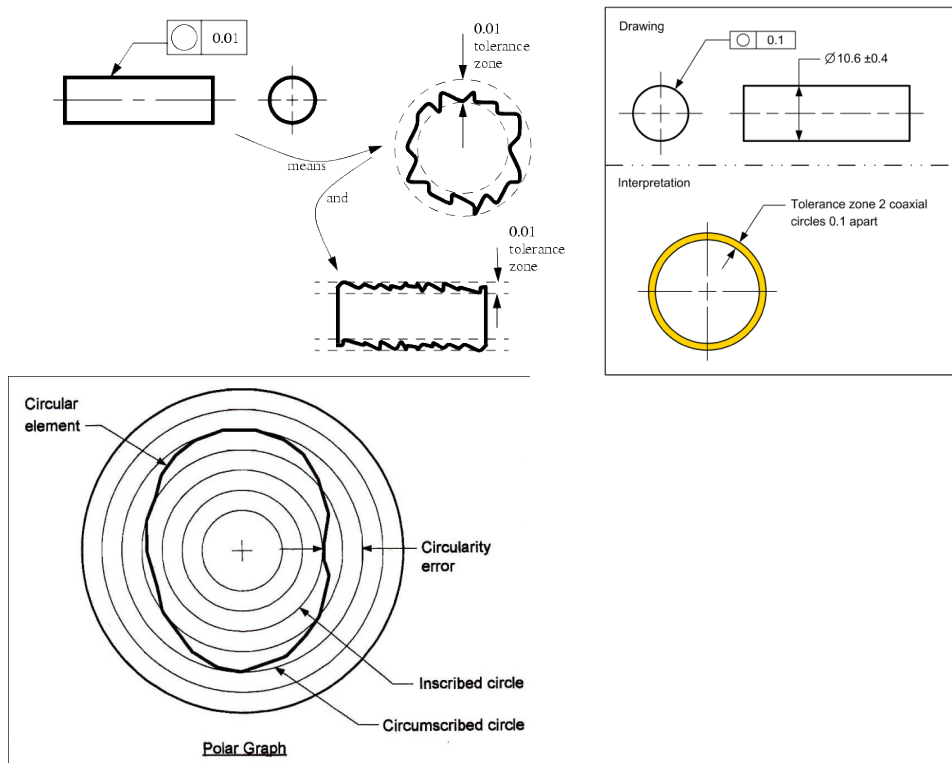
Flatness:



Straightness:

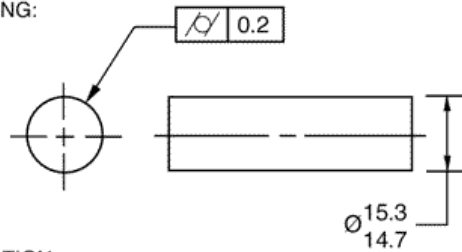


Circularity:

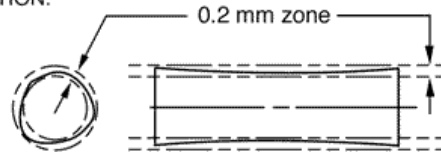


Cylindricity:

DRAWING:

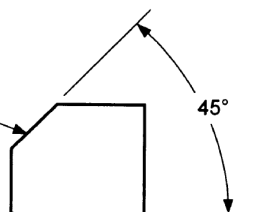
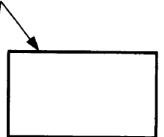
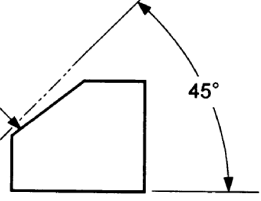
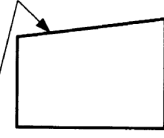


INTERPRETATION:

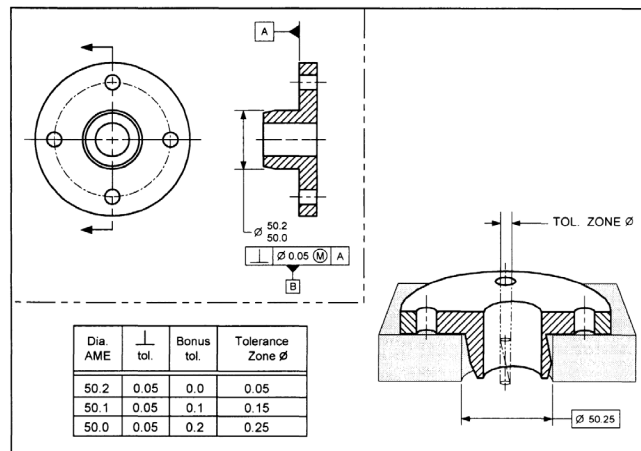
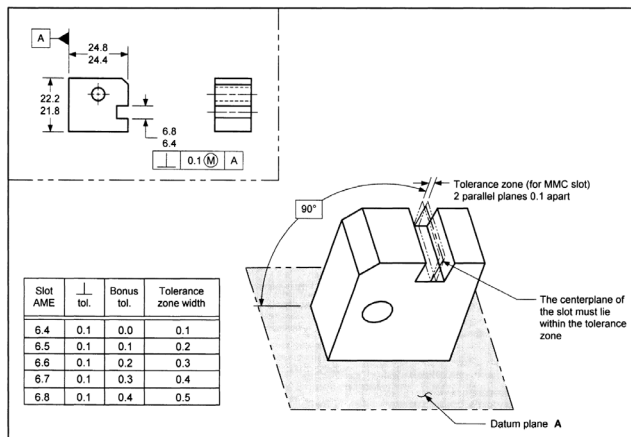
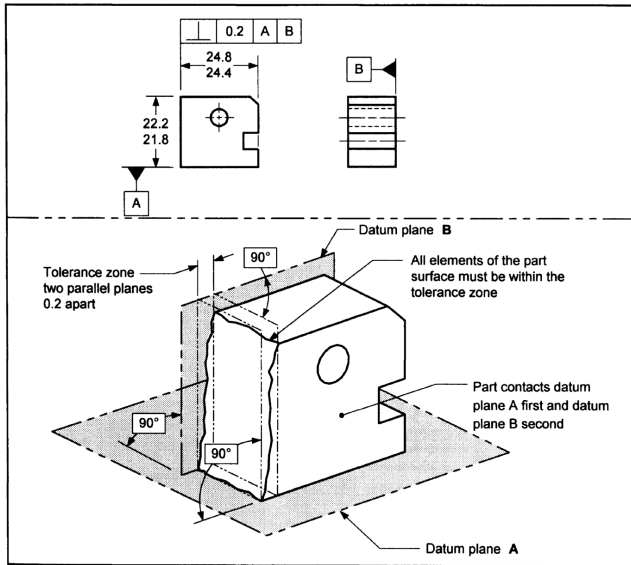


Orientation Controls:

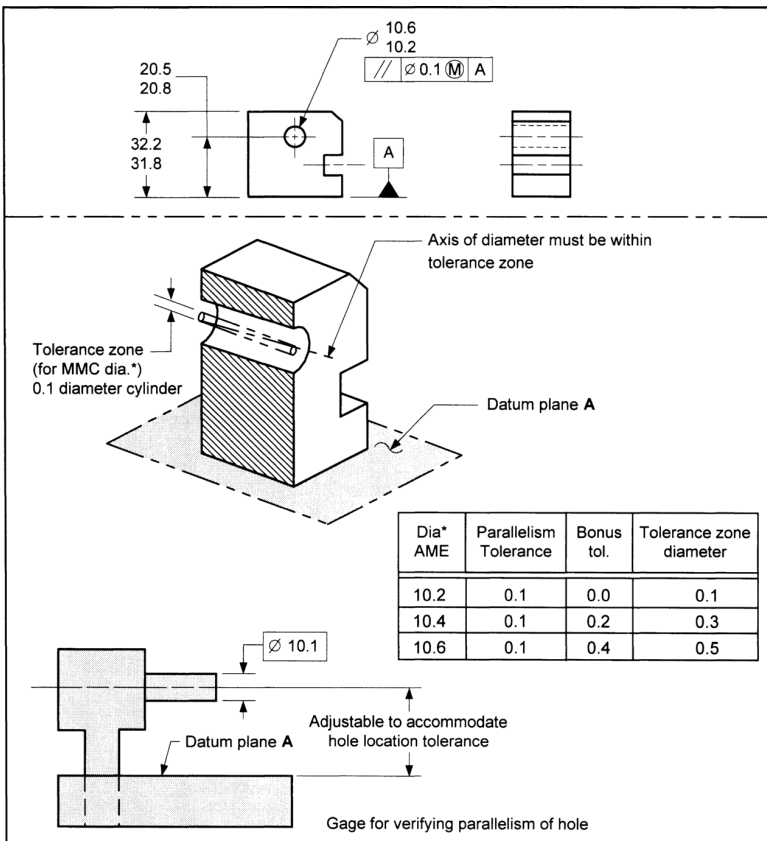
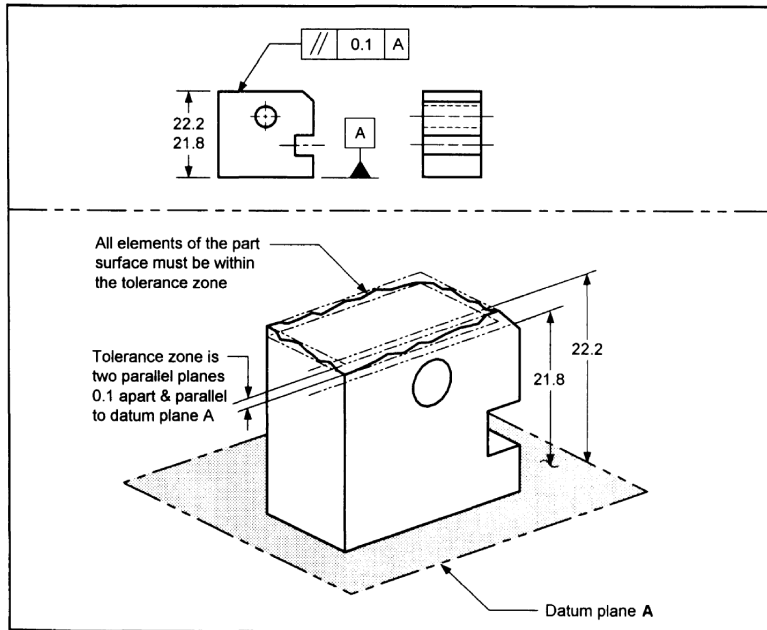
Orientation	Angularity	$\angle$
	Perpendicularity	$\perp$
	Parallelism	$//$

$\perp$ Perpendicularity	$\angle$ Angularity	$//$ Parallelism
 Perfect Perpendicularity	 Perfect Angularity	 Perfect Parallelism
 Surface with Perpendicularity Error	 Surface with Angularity Error	 Surface with Parallelism Error

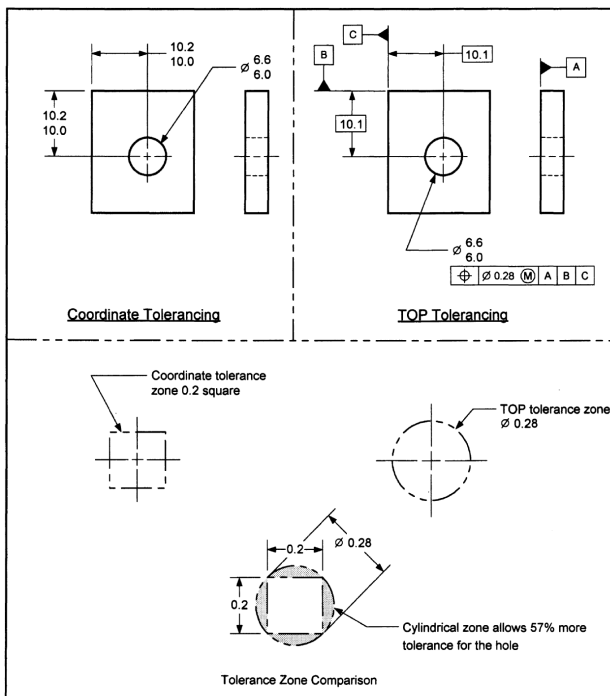
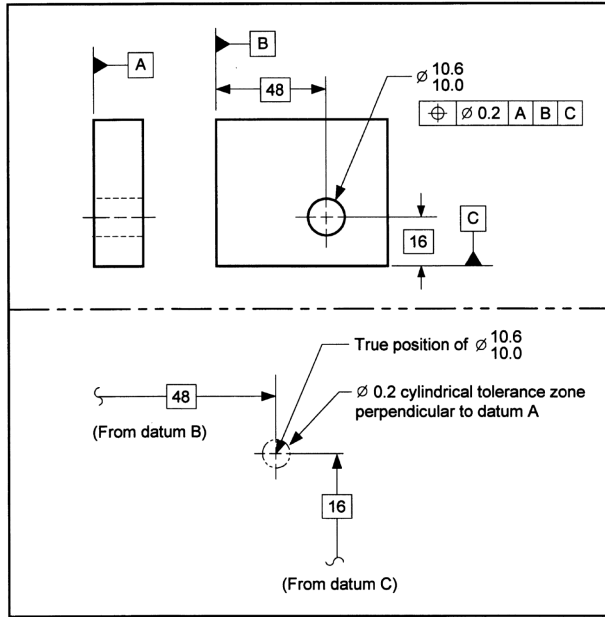
Perpendicularity control:

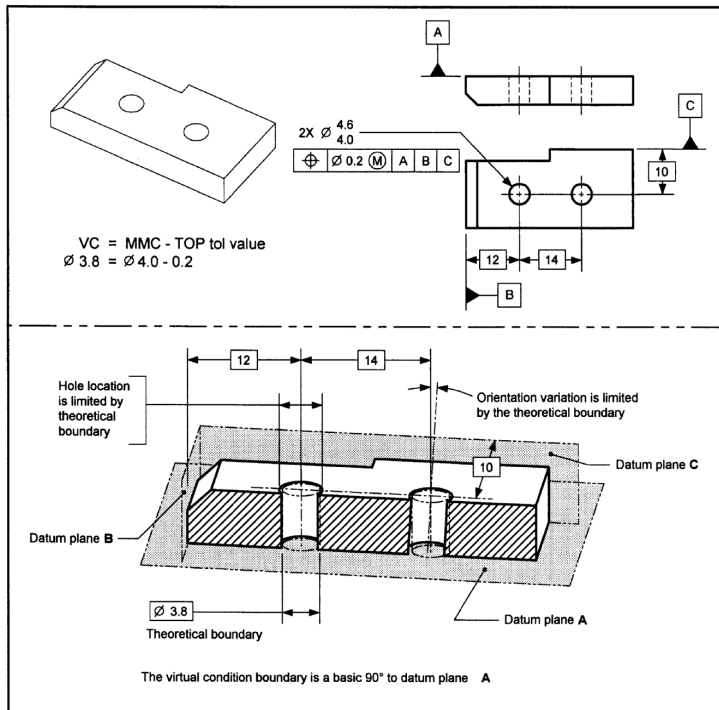


Parallelism control:

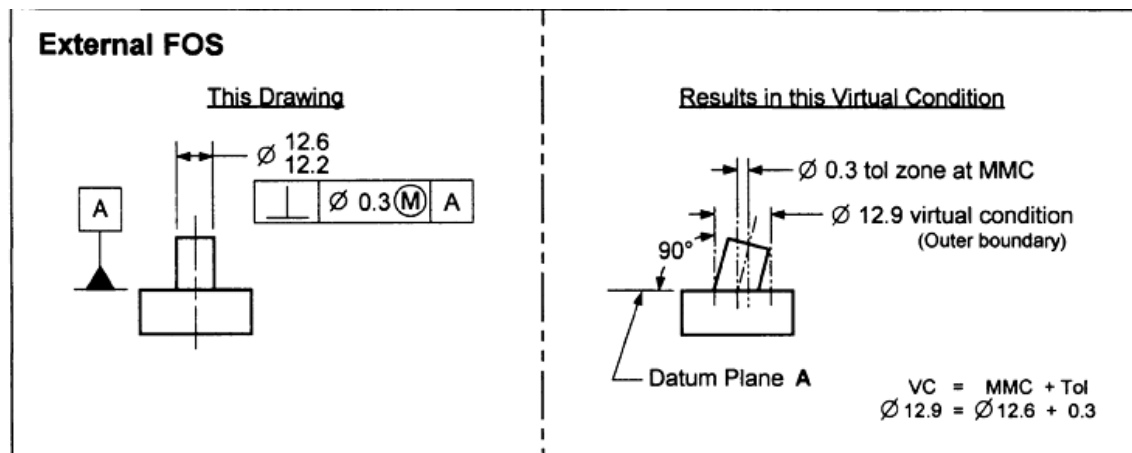
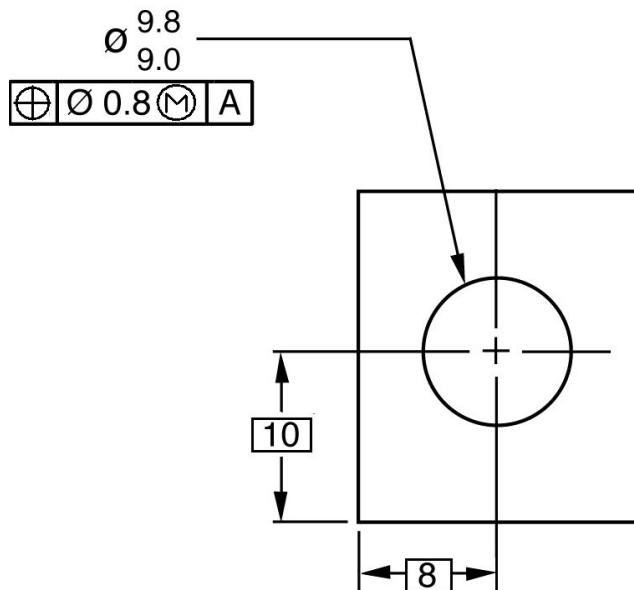


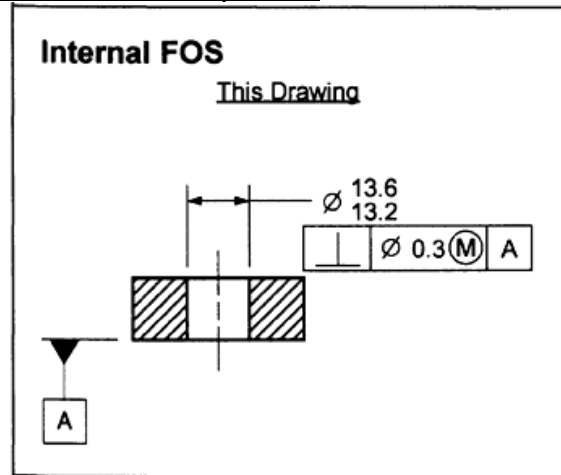
Position Tolerance:

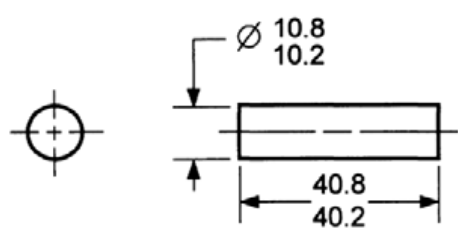
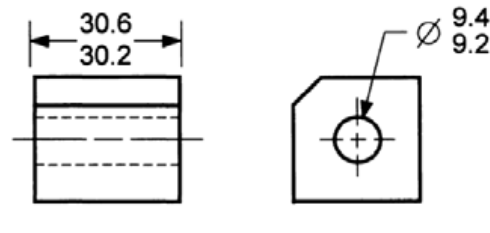
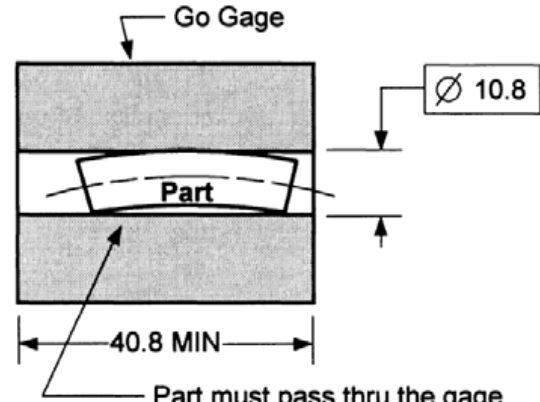
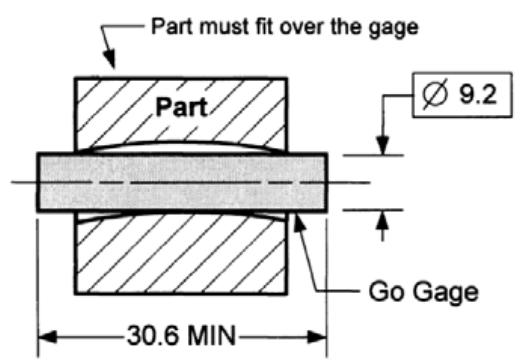
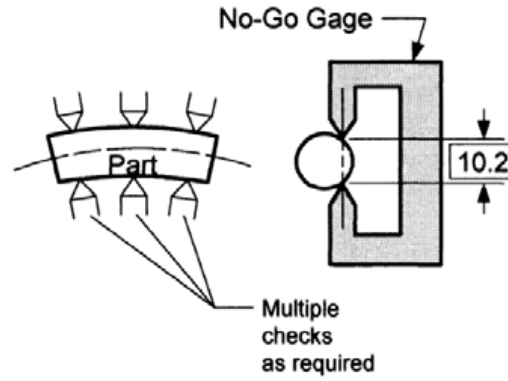
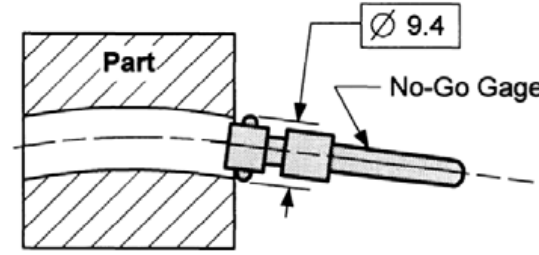




Example, Bonus Tolerance:





	
Go Gage (Verifies part diameter does not exceed MMC size and Rule #1 boundary) 	Go Gage (Verifies part diameter does not violate MMC size and Rule #1 boundary) 
No - Go Gage (Verifies that any two-point check is equal to or greater than LMC) 	No - Go Gage (Verifies that any two-point check is equal to or less than LMC)  The No-Go gage could be used at both ends of the hole. If a check inside the part is needed, a variable two-point measurement can be made.

